

# Jev, Vibe Check 🎧

## Decision Offloading

What does an agent gain  
when judgment is offloaded?



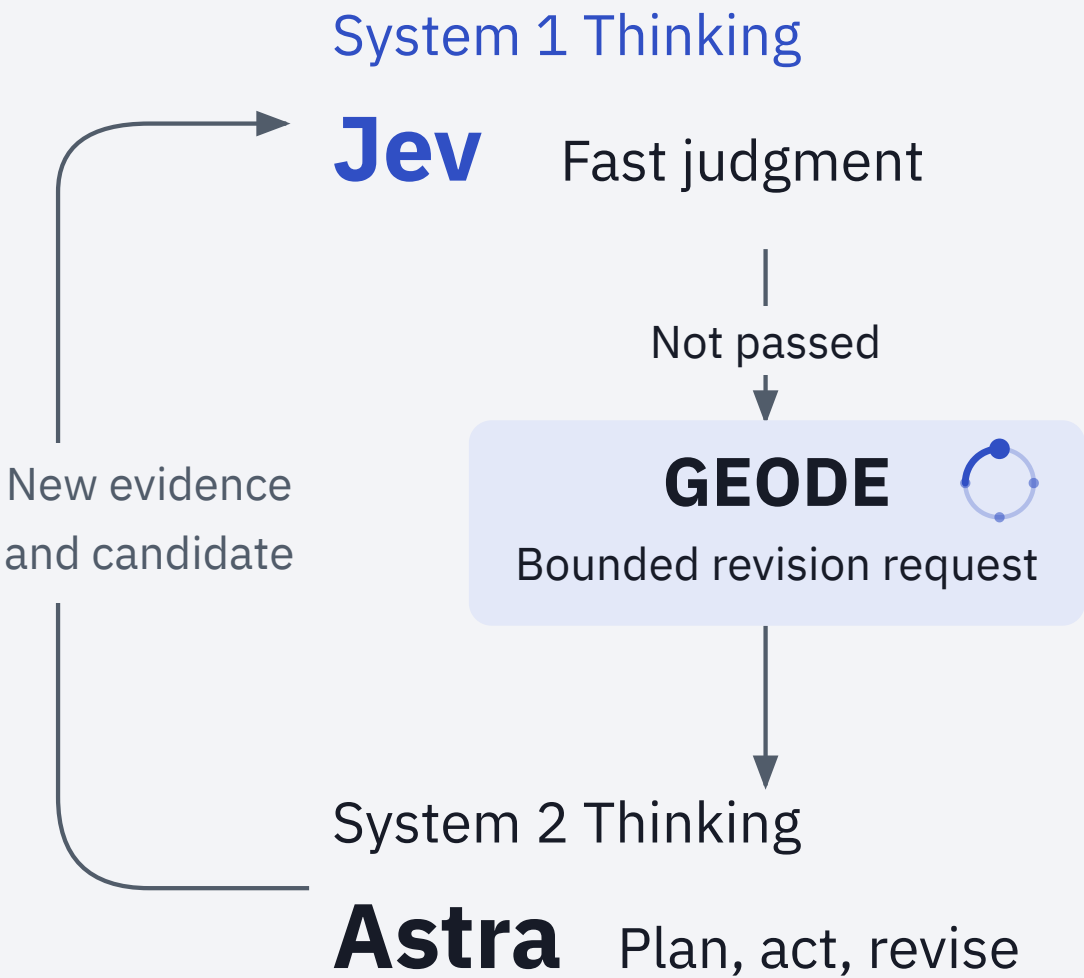
Jev

by TypeSafe AI



GEODE

Self-evolve agent harness



# Decision offloading: design and validation

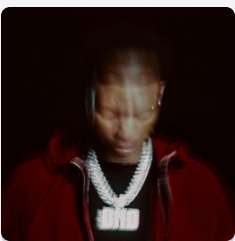
What does an agent gain by offloading its verdicts? · Korean 13:29, English 13:32

|    |                                                                                                                                                                                       |       |
|----|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|
| 01 | <b>Question / system</b><br>Jev public surface · Research question · GEODE and judge replacement                                                                                      | 13:43 |
| 02 | <b>Implementation / pilot</b><br>Development scaffold · Pilot observations · Public surface and research                                                                              | 14:54 |
| 03 | <b>Design / methods</b><br>Comparison and completion criteria · Data and generation · Procedure and tracks · Policy freeze and analysis · Metric reading guide                        | 17:02 |
| 04 | <b>Quality by task</b><br>Completion / Choice · Joint judgment / Noul · Intent and target · Candidate selection / Score · External trajectory judgment · External candidate selection | 20:30 |
| 05 | <b>Time / cost</b><br>Main study / verifier latency · Calls and accounting scope                                                                                                      | 21:38 |
| 06 | <b>Signals / policy</b><br>Repeat consistency · Error ranking and calibration · Frozen conditional offloading                                                                         | 21:59 |
| 07 | <b>Execution</b><br>Repeated runs · fault injection · repair and delivery                                                                                                             | 22:44 |
| 08 | <b>Use / reproduction</b><br>Operating conditions · Public evidence · Reader review paths · Sources and production                                                                    | 25:57 |
| 09 | <b>Conclusion</b><br>Findings and adoption                                                                                                                                            | 26:48 |

# Decision offloading: design and validation

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|    |                                                                                                                                                             |       |
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| 09 | <b>Conclusion</b>                                                                                                                                           | 26:48 |
|    | Findings and adoption                                                                                                                                       |       |



**SOUNDTRACK**  
**Nemzzz**  
LOCKED IN · drill · 136–151 BPM

|    |                                |            |
|----|--------------------------------|------------|
| 01 | <b>GASS</b> feat. Travis Scott | 140 BPM    |
|    | ■□□□ ■□□□ ■□■□ ■□■□            | 0:00–1:54  |
| 02 | <b>KILLIN EM</b>               | 144 BPM    |
|    | ■□■□ ■□■□ ■□■□ ■□■□            | 1:54–3:47  |
| 03 | <b>TODAY</b>                   | 142 BPM    |
|    | □□□□ □□■□ □□□□ ■□□□            | 3:47–5:50  |
| 04 | <b>TOKYO DRIFT</b>             | 136 BPM    |
|    | ■□■□ □□■□ □□■□ ■□□□            | 5:50–7:38  |
| 05 | <b>PRINCE OF THE SCENE</b>     | 149 BPM    |
|    | ■□□□ ■□□□ ■□□□ □□□□            | 7:38–9:34  |
| 06 | <b>4AM</b>                     | 151 BPM    |
|    | ■□□□ ■□□□ ■□□□ ■□□□            | 9:34–12:10 |

First-cycle times · measured cues, build-specific reading time  
After the six tracks, repeat from GASS

# Three primitives for decisions in code



Jev · typed judgments and probabilities

State and typed questions in; typed answers and probabilities out. Code decides the next action.

## QUESTION TYPE

- Choice · select an option
- Noul · yes or no
- Score · ordered levels

## TYPED ANSWER

choice + probabilities  
confidence

label: P(.)

...

p

$P(\text{yes}) \in [0, 1]$

$\Sigma$

$\text{score} = \Sigma i \times P(i)$

legend · probabilities · confidence

## CODE CONSUMER

- Choose a branch
- ?

Check a condition
- Rank by rubric

# Public API, undisclosed internals

## Documented API contract

POST /v1/systemone

|                  |                                     |
|------------------|-------------------------------------|
| <b>model</b>     | Model identifier                    |
| <b>state</b>     | Request, candidate and observations |
| <b>questions</b> | Named questions and answer types    |

## Returned values

A typed answer for each question

Independent questions share one state in one call.

An answer-dependent question needs a later call.

## Limits of the public contract

VENDOR ACCOUNT

**New architecture · parallel sampler**

RLCD

Launch and founder claims, not an open implementation

NOT VERIFIED IN REVIEWED SOURCES

**Backbone · attention mask · loss · weights**

An encoder or decoder reused from a specific LLM?

Possible, but not established by the public contract.

## The caller still decides: acceptance threshold, next action, retry / repair budget

This study tests returned judgments and subsequent GEODE actions, not Jev's internal architecture.

# When can a faster judgment be delegated?

Final native evidence · editorial interpretation

Original runs preserved; corrections labeled separately

Offloading depended on the judgment and delivery contract.

Fixed intent classification

Does it identify the correct intent and target?  
Separate panel items from actual tasks

Noul joint · candidate selection

Does quality hold for joint conditions and candidate ranking?  
Each primitive has its own comparison unit

Probability · conditional offloading

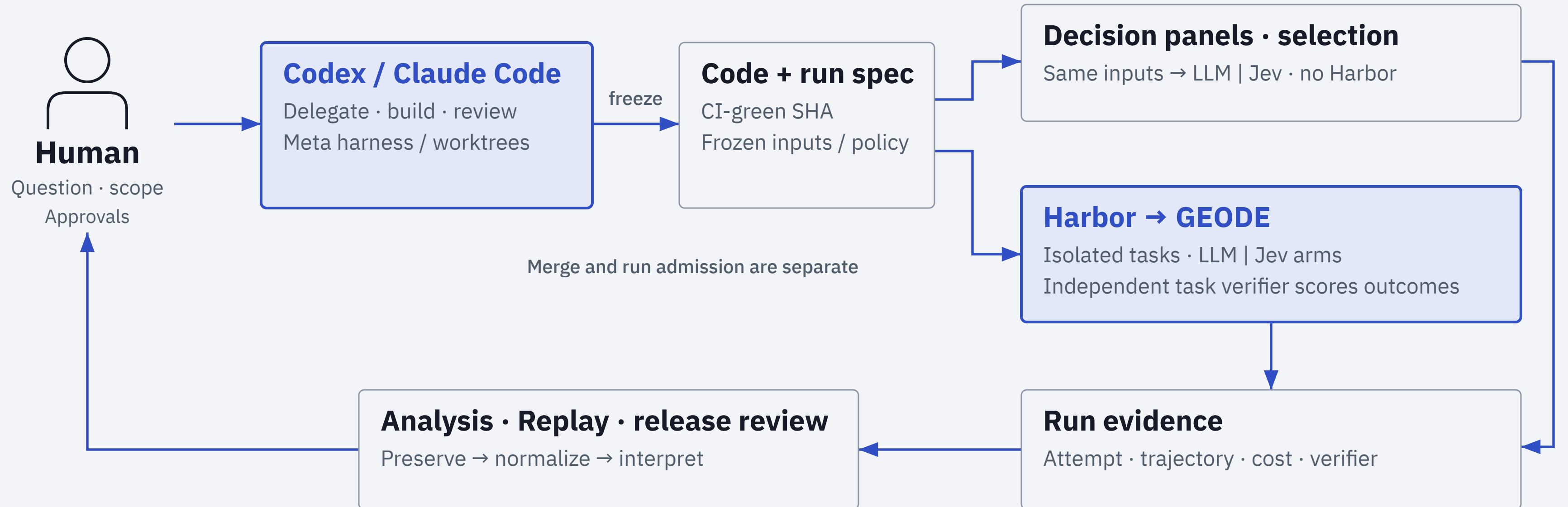
Can uncertain judgments go to a stronger judge?  
Freeze policy; evaluate panel and execution separately

Next: matched-input quality, signals that rank errors, then actual delivery.

# Who builds it: and what proves it works?

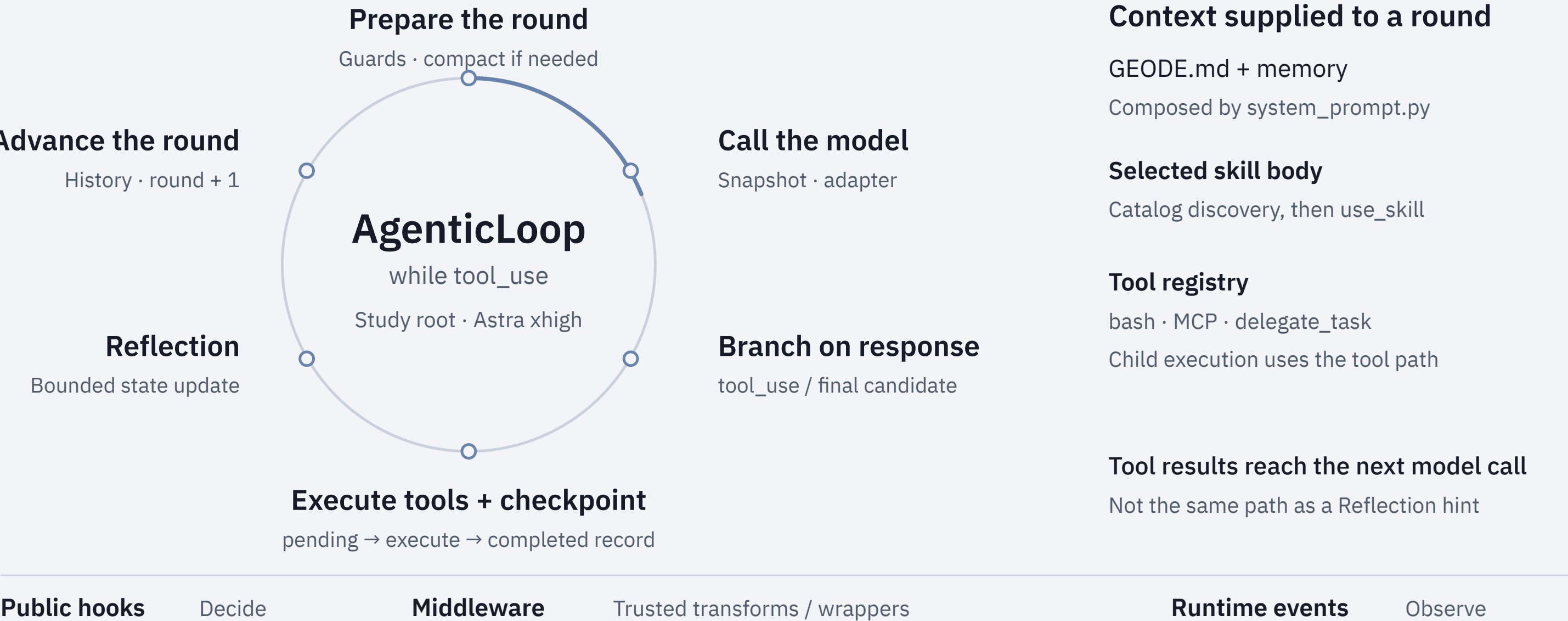
From a human question to verifiable evidence

Coding hosts ≠ experimental arms

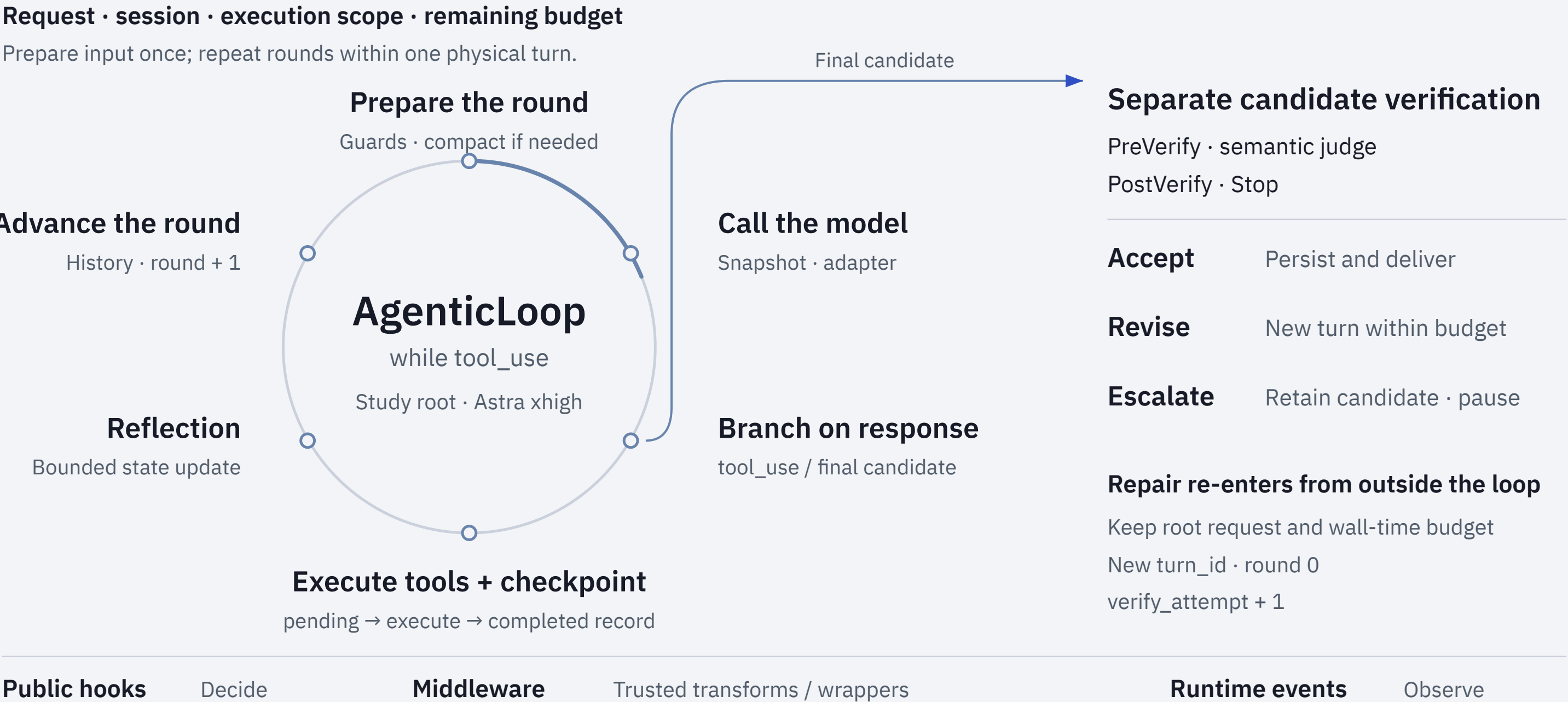


# GEODE’s execution map: rounds and final verification

**Request · session · execution scope · remaining budget**  
Prepare input once; repeat rounds within one physical turn.



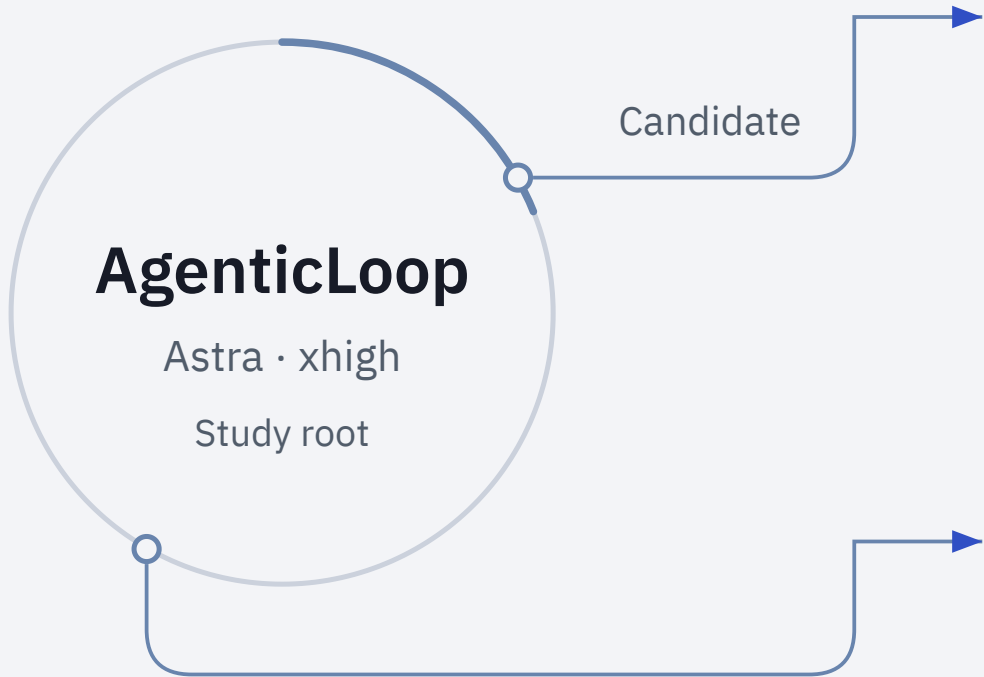
# GEODE's execution map: rounds and final verification



# Where Jev fits: judgments remain separate from actions

## Keep the main execution loop

Current core Jev routes require explicit selection



## Choice · completion verdict

|           |                                              |
|-----------|----------------------------------------------|
| Input     | Root request · candidate · retained evidence |
| GEODE map | passed · should_retry · feedback             |
| Reader    | Finalizer · deliver / repair / escalate      |

GEODE retains execution authority and repair budgets

## Choice · evidence classification

|        |                                            |
|--------|--------------------------------------------|
| Input  | Observations + bounded cognitive state     |
| Write  | Append a code-owned next check to subgoals |
| Reader | Next Reflection · state snapshots          |

Not belief/confidence synthesis or direct root-prompt injection

## Reflection after the tool checkpoint

### Evaluation-only · opt-in

### Choice intent + target classification

analyze\_request returns to root context; no action execution or permission

# Where Jev fits: judgments remain separate from actions

## Keep the main execution loop

Current core Jev routes require explicit selection



## Comparisons at design time, separate from main paths

### Score

Selection quality on the same four frozen candidates

Current candidate\_sampling selection uses an LLM

### Noul

Judge contradictions and missing evidence separately

Measure joint judgments and subsequent repair

### Cascade

Escalate only uncertain Jev judgments to Astra

Compare thresholds, call cost and false acceptance

## Keep global LLM judgment; swap only assigned slots

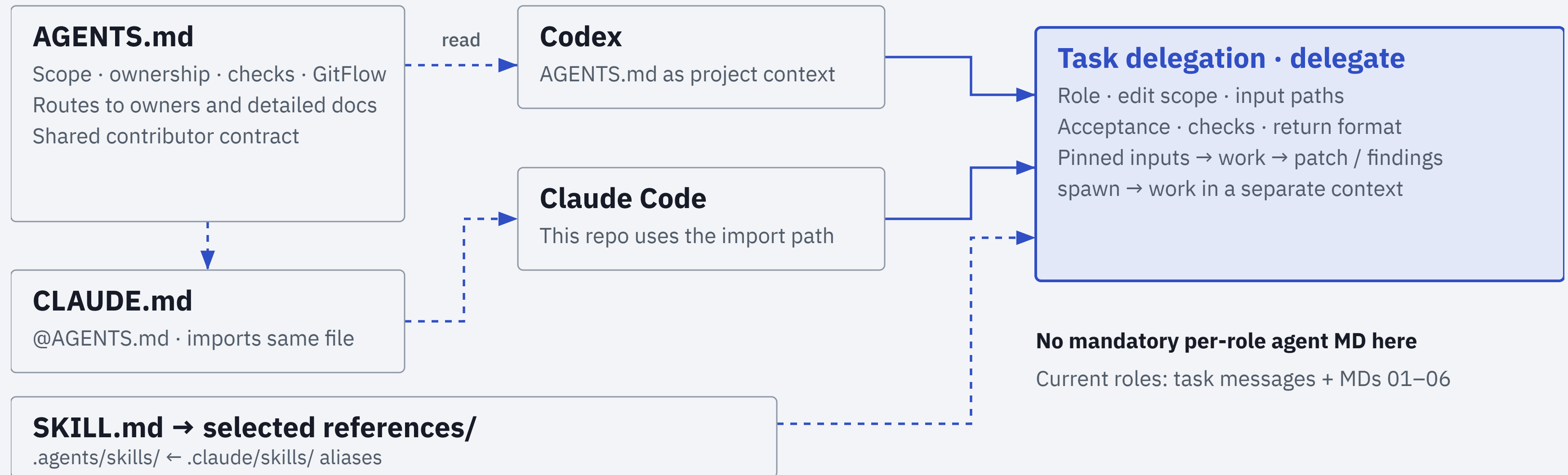
### Evaluation-only · opt-in

### Choice intent + target classification

analyze\_request returns to root context; no action execution or permission

# One contract, two coding-host contexts

HOST · System instructions / tools / permissions and sandbox



# AGENTS.md: a map to the right instructions

## From shared rules to relevant evidence

### AGENTS.md

Scope · ownership · acceptance · verification  
Codex reads it; CLAUDE.md imports @AGENTS.md

- docs/workflow.md → task owner  
Route the request to its code and document owner
- .agents/skills/ → SKILL.md  
.claude/skills/ aliases the same source files
- Selected references → code, callers and tests  
Read the chosen skill fully, then follow the evidence

## Routes selected for this project

### Design and measurement

[geode-eval](#) · [stanford-test-time-compute](#)  
05 preregistration + 06 reliability → freeze / admission

### Build and integration

[geode-workflow](#) · [geode-gitflow](#)  
Bounded role contracts → patches → owner verification

### Film and review

[technical-deck-report](#) · [viz-frame-audit](#)  
DESIGN.md + fluent-korean → KO/EN render review

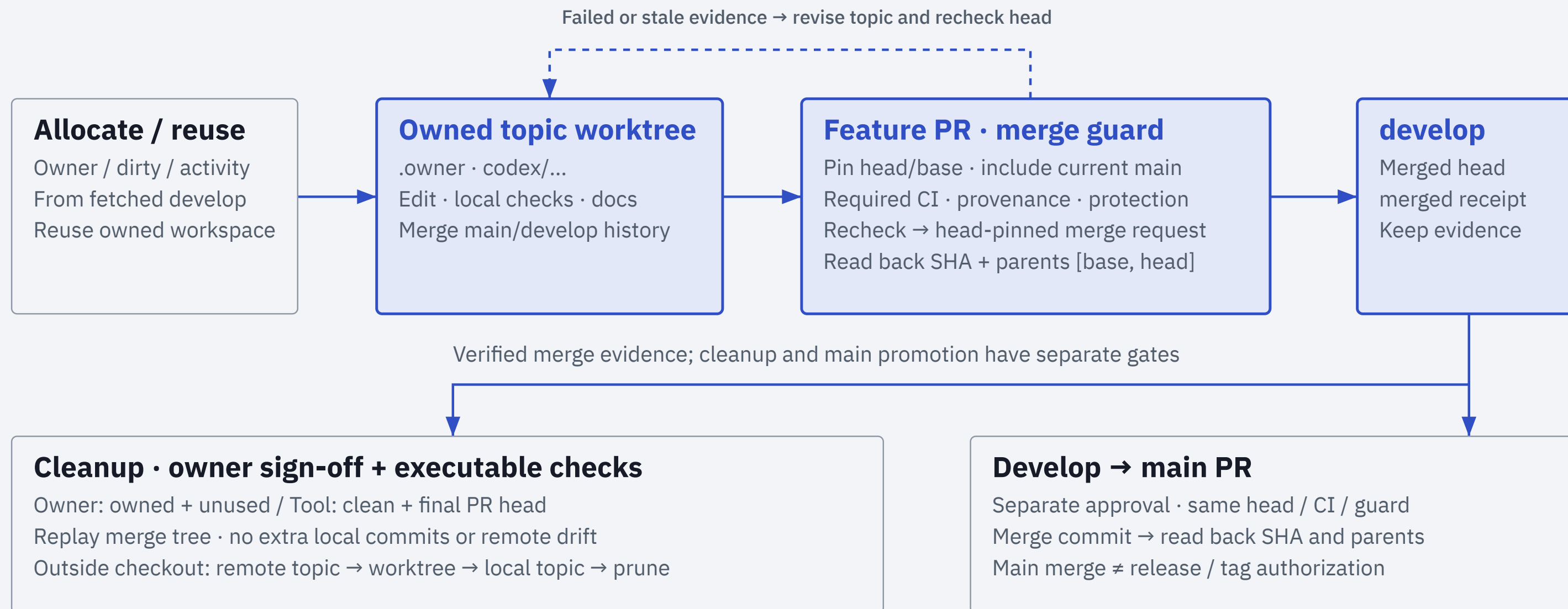
- Delegation

Role, inputs, edit scope, acceptance → spawn → patch / findings
- Iteration

Owner reproduces and checks → findings back → revise and recheck

GEODE.md + .geode/skills/ supply runtime context; the host enforces permissions and sandboxing.

# Worktree lifecycle, from isolation to merge and cleanup



Uncertain merge outcome → verify remote state before retry or cleanup

**ADMISSION** Freeze a verified SHA, inputs and policy; pass admission to run. Main promotion follows separate approval.

# A faster verdict, but why did one task take longer?

○ LLM ■ Jev One line = one matched task–repetition pair

Final-judge call · seconds

explicit · r0

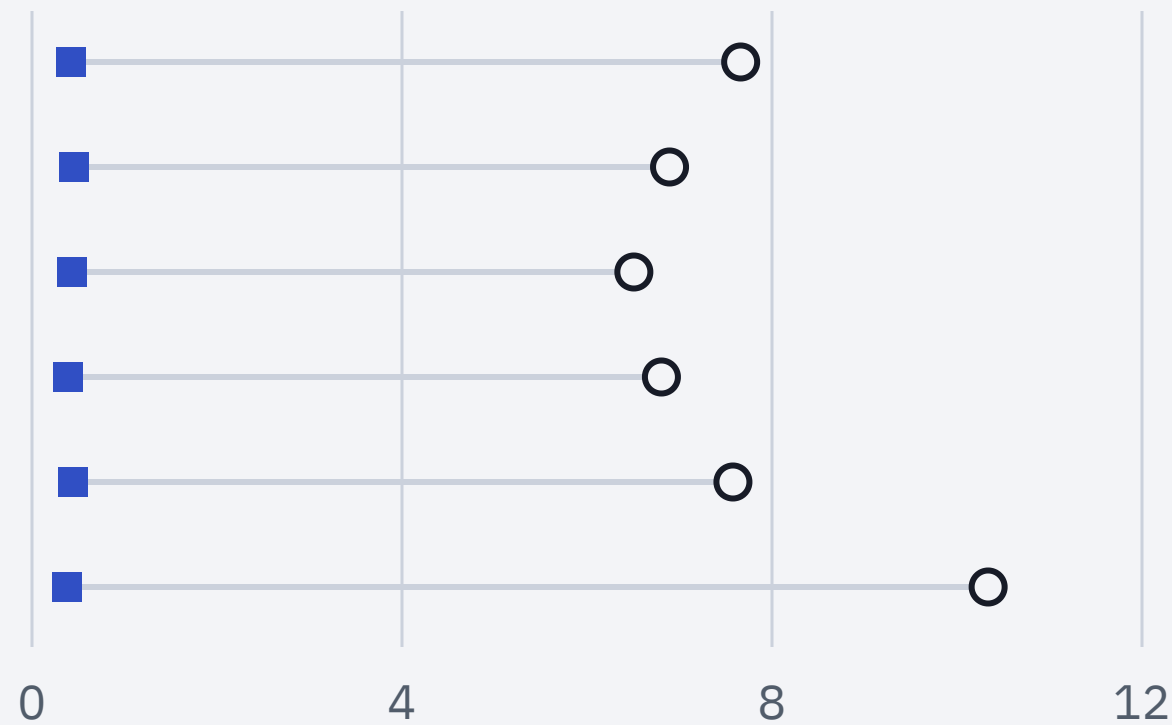
explicit · r1

context · r0 \*

context · r1

korean · r0

korean · r1



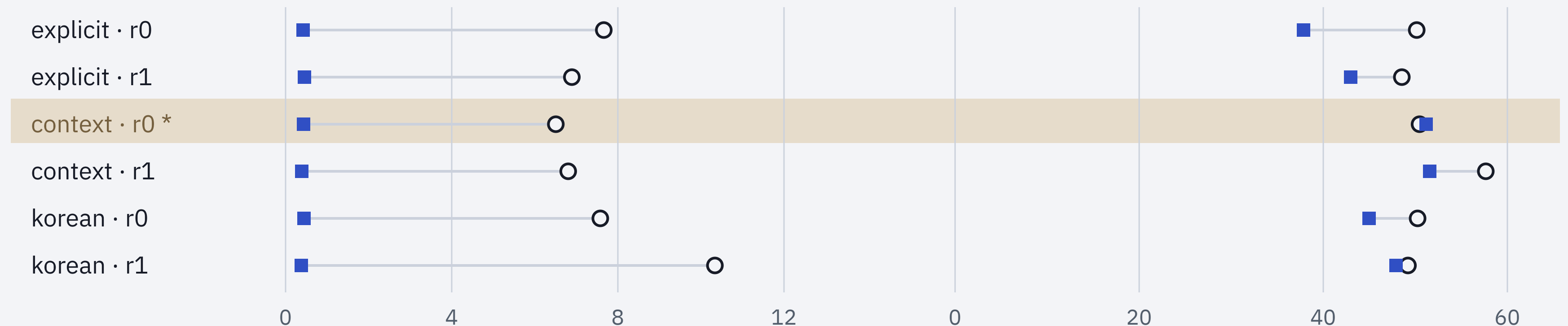
Judge median: 7.235 s → 0.426 s

# A faster verdict, but why did one task take longer?

○ LLM ■ Jev One line = one matched task–repetition pair

Final-judge call · seconds

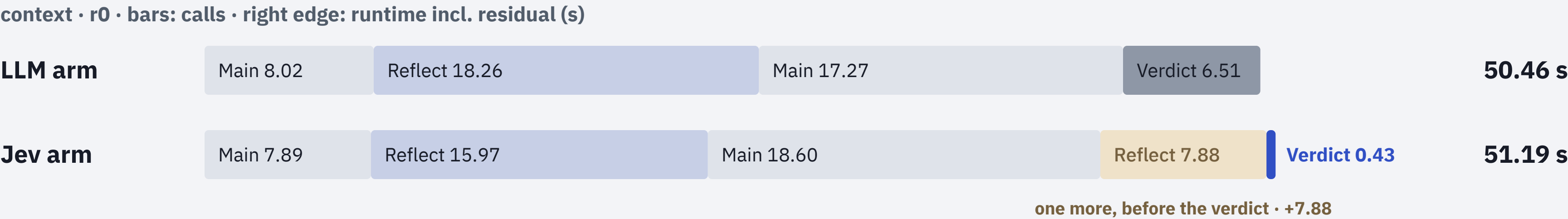
GEODE runtime · s (setup and oracle excluded)



Judge median: 7.235 s → 0.426 s

Runtime median: 50.198 → 46.439 s · both 6/6 pass

# A faster verdict, but why did one task take longer?



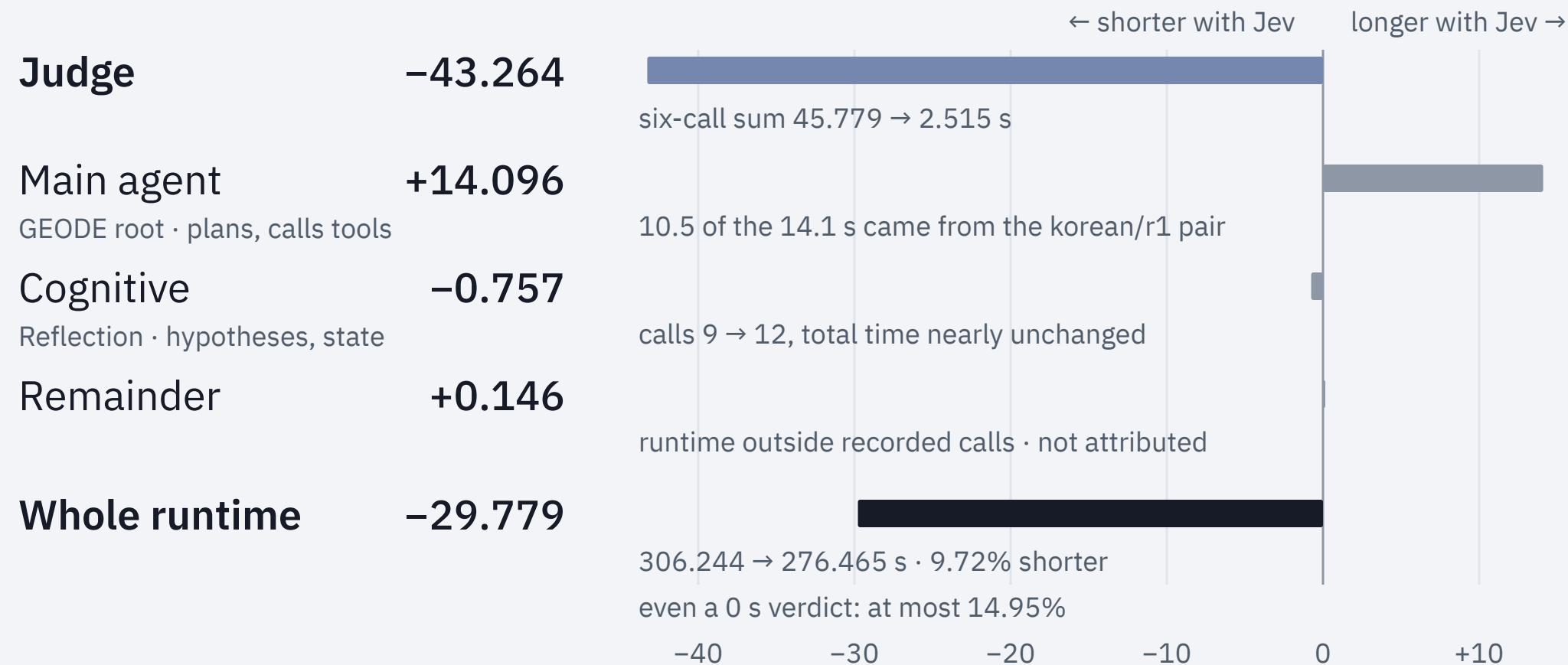
verdict -6.07 · extra reflection +7.88 · first reflection -2.29 · main +1.21 · rest +0.01 → runtime +0.74 s

One more Cognitive reflection ran right before the verdict: its 7.88 s outweighed the 6.07 s the verdict saved.

Cadence: Astra Cognitive confidence ≥ 0.8 doubles the interval. Logged: LLM 0.8; Jev 0.65 after the extra pass. First-pass confidence was not logged; the cause of the extra pass remains unresolved.

# Six verdicts: 18.2× speedup, what about total time?

Six-pair sums · Jev – LLM · seconds · Natural runs r2 natural (3 inboxes × 2 repeats)



One verdict · explicit/r0

**7.661  $\rightarrow$  0.419 s**

×18.3

Astra API-equiv. \$0.0398 · Jev list price \$0.00011

## Six verdicts

45.779 → 2.515 s

×18.2

\$0.2448 · \$0.00065 · 14.95% of baseline time

## Whole task · six pairs

306.244 → 276.465 s

**×1.11**

Astra-equiv. \$1.077  $\rightarrow$  \$0.900 .

Jev separate · billing unknown

Holding other stages fixed: a zero-time verdict caps savings at  $14.95\% \cdot \text{ratio} = \text{LLM} \div \text{Jev}$

Verdicts -43.264 s + other stages +13.485 s = total -29.779 s (-9.72%).

Call order: in all twelve runs its calls finished before the verdict call, and no request went back after the verdict.

The increase was observed in Astra execution before the Jev call:

10.5 s from korean/r1 (first call, same 3,011 input tokens: 9.6 → 14.3 s; both calls' output 512 → 723 tokens)

# The same answer, different observations?

The request and candidate “Order A-104 has shipped” stay fixed across three tool observations.

| Supplied observation                                                   | LLM   Jev · actual verdict                                                                  | Code-projected branch                    |
|------------------------------------------------------------------------|---------------------------------------------------------------------------------------------|------------------------------------------|
| <div>status: shipped</div> <div>Supports the same candidate</div>      | <div>supported   supported</div> <div>Jev confidence 1.00</div>                             | <div>Pass → no revision</div>            |
| <div>status: cancelled</div> <div>Contradicts the same candidate</div> | <div>contradicted   contradicted</div> <div>Jev confidence 1.00</div>                       | <div>Not passed → request revision</div> |
| <div>observations: []</div> <div>No evidence for the candidate</div>   | <div>insufficient_evidence<br/>  insufficient_evidence</div> <div>Jev confidence 0.95</div> | <div>Not passed → request revision</div> |

SCOPE

Both engines matched 3/3 labels. Repair branches are code projections; this diagnostic made no downstream agent call.

# Missing evidence, a “contradiction” verdict

Same input · A-104 candidate: shipped · Lookup observations: 0

## LLM Astra / xhigh

Request contract · one field: verdict

Actual return

```
{"verdict": "insufficient_evidence"}
```

Response latency 6.607 s · 929 input / 106 output tokens

...step-2-025475da

## Jev Jev 1.13

Request contract · Choice primitive

Actual return

```
{"verdict": {  
  "type": "choice", "choice": "contradicted",  
  "confidence": 0.58,  
  "probabilities": {"supported": 0.0,  
    "contradicted": 0.72,  
    "insufficient_evidence": 0.28}  
}}
```

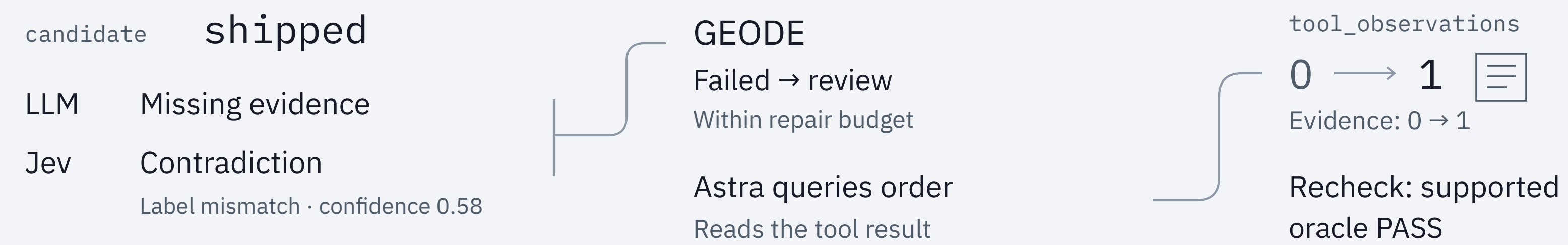
Response latency 8.506 s · 933 input / 51 output tokens

...step-2-a1a1d069

**CONSEQUENCE** Jev returned contradicted instead of insufficient\_evidence. Both labels routed the run to repair.

# The cost from a failed verdict to recovery

Before lookup: the answer was shipped, with no recorded evidence.



Answer: shipped → shipped · The new lookup supports the same answer.

| Recovery after the first negative verdict  | LLM                 | Jev                        |
|--------------------------------------------|---------------------|----------------------------|
| Replan + main agent + Cognitive + rejudge  | 1 + 2 + 2 + 1       | 1 + 2 + 2 + 1              |
| Until the final verdict completes          | 58.855 s            | 50.321 s                   |
| Recorded input / output tokens             | 8,003 / 1,405       | 8,030 / 1,363              |
| Astra API equivalent / Jev tariff estimate | \$0.150280 / no Jev | \$0.135560 / \$0.000044058 |

# Had we read the confidence, what would it have flagged?

All 13 recorded Jev verdicts, ordered by confidence. Execution used labels; confidence was logged only.

- agrees with the independent check
- miss

## Natural task runs

task verifier · 6

## Fixed-input judgments

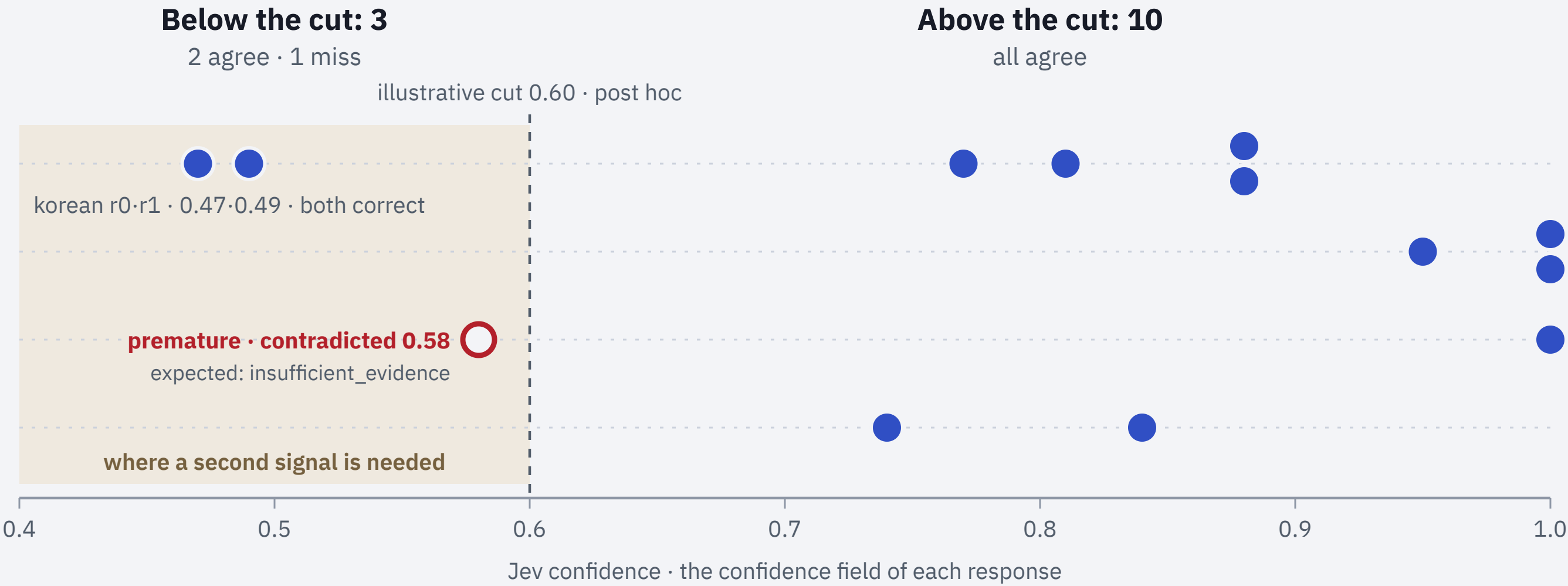
expected label · 3

## Fault injection first verdict

expected label · 2

## Fault injection re-judgment

oracle · 2



POST HOC      A cut below 0.60 would flag the one miss and two correct verdicts: 3 rechecks for 1 error.  
Threshold selection still needs held-out evidence.

# Check order A-104 status · Choice

LLM verdict | Astra | 1x | 588a1c8 | replay 00:00:00.000

[reconstructed] Harbor ATIF replay; not a raw PTY capture.  
[task] Process every inbox item.  
<inbox>[{"id": "status\_one", "request": "What is the current status of A-104?", "candidates": ["A-104"]}]\n

POST-RUN ORACLE: PASS | separate verifier receipt

Jev verdict | Astra | 1x | 588a1c8 | replay 00:00:00.000

[reconstructed] Harbor ATIF replay; not a raw PTY capture.  
[task] Process every inbox item.  
<inbox>[{"id": "status\_one", "request": "What is the current status of A-104?", "candidates": ["A-104"]}]\n

POST-RUN ORACLE: PASS | separate verifier receipt

Check order A-104 status · Choice

LLM verdict | Astra | 1x | 588a1c8 | replay 00:01:10.267END

Jev verdict | Astra | 1x | 588a1c8 | replay 00:00:50.133END / held

```
[reconstructed] Harbor ATIF replay; not a raw PTY capture.
[task] Process every inbox item.
<inbox>[{"id": "status_one", "request": "What is the current status of A-104?", "candidates": ["A-104"]}]/>
$ [lookup_order_status] {"items": [{"id": "status_one", "order_id": "A-104"}]}
{"result": {"items": [{"id": "status_one", "order_id": "A-104", "status": "shipped"}]}}
[agent] {"items": [{"id": "status_one", "order_id": "A-104", "status": "cancelled", "disposition": "answered"}]}
[agent] {"items": [{"order_id": "A-104", "status": "shipped", "disposition": "answered", "id": "status_one"}]}
█
```

```
[reconstructed] Harbor ATIF replay; not a raw PTY capture.
[task] Process every inbox item.
<inbox>[{"id": "status_one", "request": "What is the current status of A-104?", "candidates": ["A-104"]}]/>
$ [lookup_order_status] {"items": [{"id": "status_one", "order_id": "A-104"}]}
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[agent] {"items": [{"order_id": "A-104", "status": "shipped", "disposition": "answered", "id": "status_one"}]}
█
```

CONTROLLED INPUT: {"items": [{"id": "status\_one", "order\_id": "A-104", "status": "cancelled", "disposition": "answered"}]}

CONTROLLED INPUT: {"items": [{"id": "status\_one", "order\_id": "A-104", "status": "cancelled", "disposition": "answered"}]}

Judgment 2: supported | 5311.9 ms

Judgment 2: supported | 468.7 ms

CODE FEEDBACK: Confirm the final response retains those evidence boundaries.

CODE FEEDBACK: Confirm the final response retains those evidence boundaries.

Feedback consumed: step-2-53687a5a (confirmed at call end)

Feedback consumed: step-2-2f43ab05 (confirmed at call end)

POST-RUN ORACLE: PASS | separate verifier receipt

POST-RUN ORACLE: PASS | separate verifier receipt

Replay ended · reading the results

# How far can a verdict probability be trusted?

## RLHF

Human preference between responses

**Goal: Preferred responses**

## RLVR

Rewards verified by answers and tests

**Goal: Passing verifiable checks**

## RLCD

Training signal not disclosed

**Goal: Calibrated decision probabilities**

RLCD's stated goal: align reported probabilities with observed frequencies.

# How far can a verdict probability be trusted?

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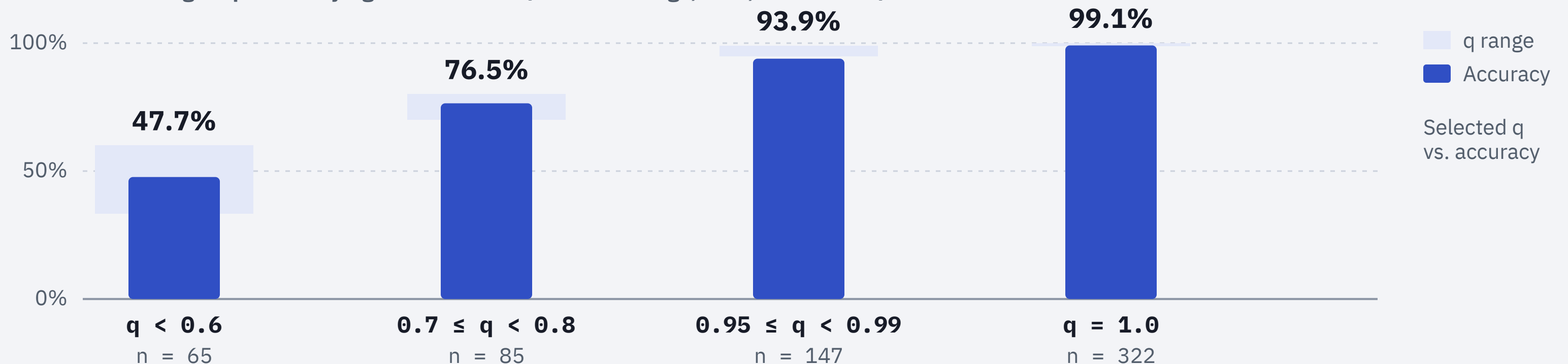
## RLCD

Training signal not disclosed

**Goal: Calibrated decision probabilities**

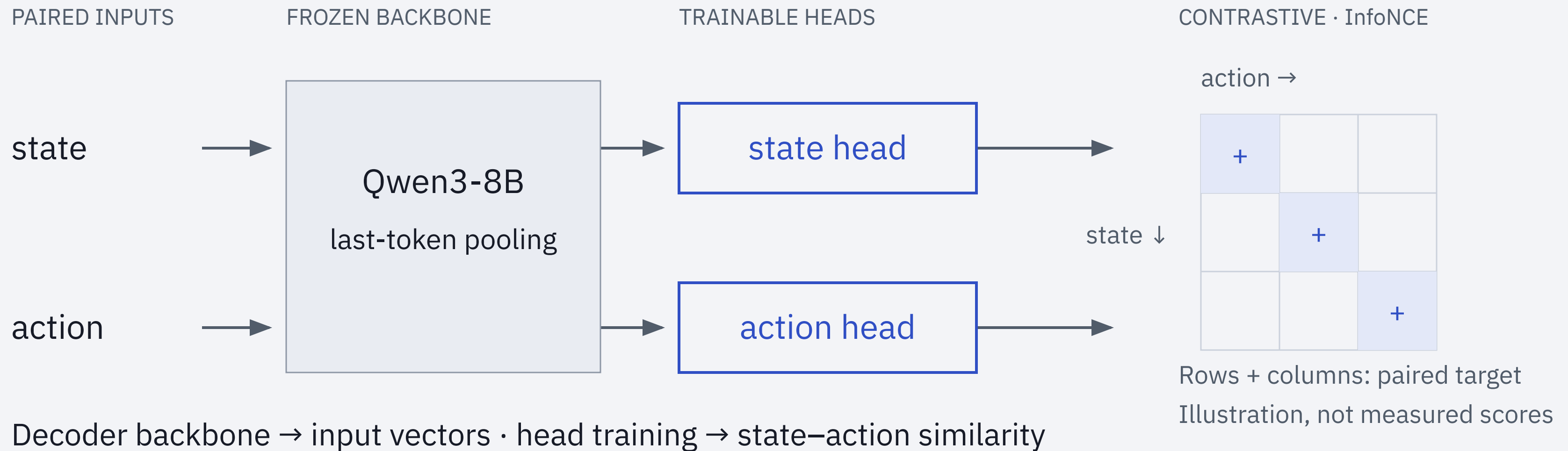
RLCD's stated goal: align reported probabilities with observed frequencies.

External measurement · was a higher probability right more often? (JEV-as-a-Judge, CMU, 2026-09-22)



External-paper measurements; GEODE calibration and error ranking were evaluated separately.

# CLM: contrastive learning for state–action pairs



## TRAINING

The public trainer targets recorded state–action pairs. Whether high similarity yields task success requires a separate evaluation.

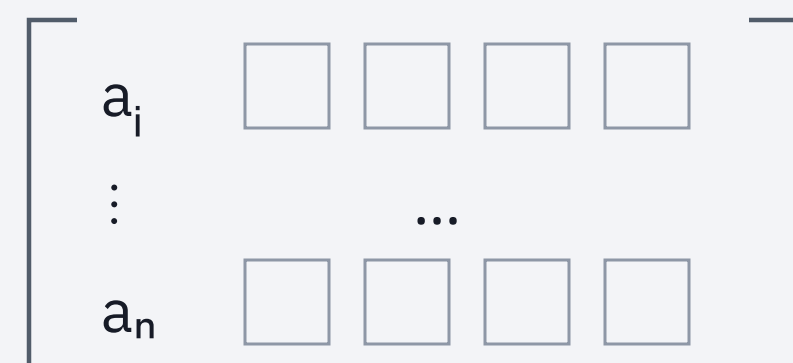
# Reusable action vectors for recurring candidates

PREPARE ONCE

Fixed candidates

Qwen → action head

Reuse vectors



scaled cosine

$\cos(s, a_i)$

$\vdots$

$\cos(s, a_n)$

→ softmax

Candidate probs.

$P(a_i | \text{pool})$

...

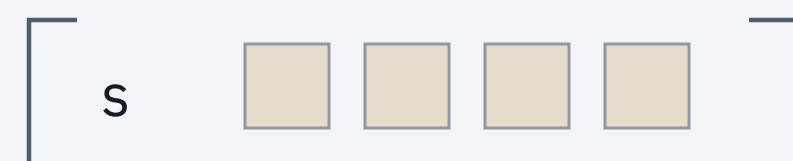
Within the supplied pool

EACH DECISION

New state + question

Qwen → state head

New vector



Compare each candidate with s

Reuse requires stable candidate text, preprocessing, backbone and heads.

COST SCOPE

The author's timing covers scoring with cached candidates. GEODE evaluation must also count candidate preparation, repeated scoring and wrong-branch repair.

Author's warm-cache result: 44 ms / 579 ms at 1,024 candidates · local CLM / hosted Jev

200-token state · 15 tokens/candidate · RTX 4090 · 5 trials · excludes initial candidate preparation

# Change the judgment model, retain the execution owner

Comparison design / fixed-input evaluation and live execution

What changed, and what stayed fixed?

## Separate a judgment on fixed input from the execution that consumes it

Changed

**Judgment model**

○ **A · GPT-6 Astra**

Codex subscription · effort xhigh

■ **B · Jev 1.13.0**

Direct API · typed responses

**Fixed-input comparison**

Matched states, pools or requests; the same task-specific criteria

Source, spec, input, policy and budget pinned before each run

**E2E execution owner**

**Astra xhigh / GEODE in both arms**

The root uses tools and repairs; intermediate observations and paths can differ.

**Task-success evidence**

Separate task verification plus execution-path and observation records

Judge acceptance  $\neq$  task success; provider prompts are not byte-identical

# The adopter defines completion; verifiers check it

Adopter design guide / this diagnostic contract

What observable evidence counts as completion?

## Define the completed work, then make it observable

| Adopter decision                                                            | Evidence that tests the decision                     | How to design the criterion            |
|-----------------------------------------------------------------------------|------------------------------------------------------|----------------------------------------|
| <ul style="list-style-type: none"><li>Completed deliverable</li></ul>       | What must be returned, saved or changed?             | Write a contract from real valid cases |
| <ul style="list-style-type: none"><li>Unacceptable errors</li></ul>         | Wrong targets, false completion, unauthorized change | Set withhold / reject rules by risk    |
| <ul style="list-style-type: none"><li>Success and resource limits</li></ul> | Verifier, time, cost and retry records               | Test boundary cases; freeze thresholds |

A design guide, not a recommendation to adopt this study’s thresholds as a business SLO.

# The adopter defines completion; verifiers check it

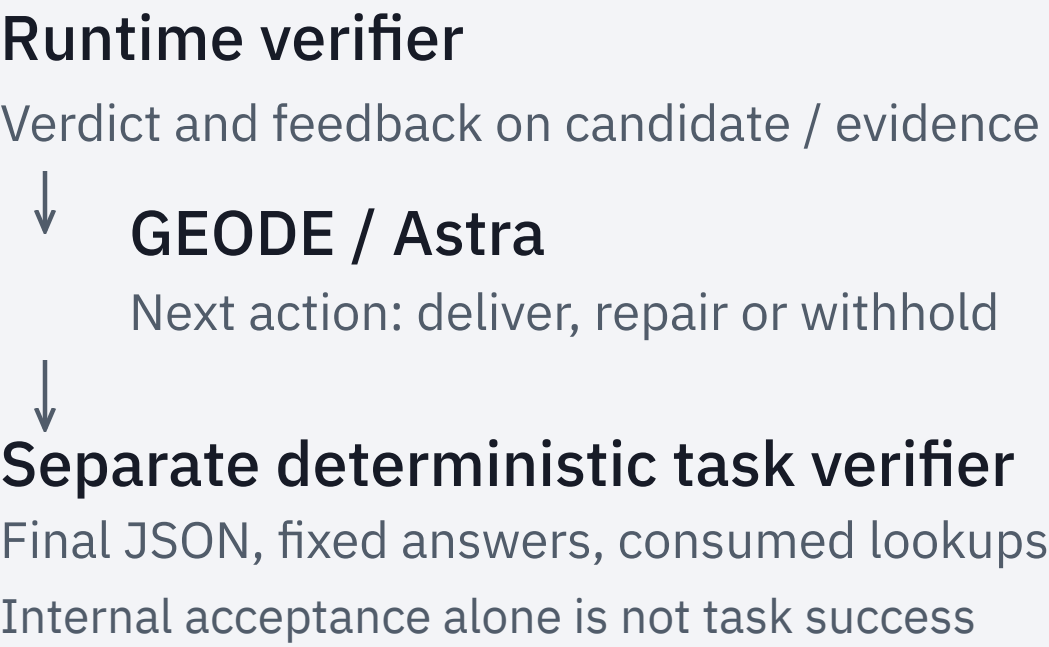
Adopter design guide / this diagnostic contract

What observable evidence counts as completion?

## Our task: return final JSON for every read-only inbox item

A fixed synthetic sandbox contract, not customer delivery or an order mutation

|                 |                                        |
|-----------------|----------------------------------------|
| Status request  | Look up the target; consume its status |
| Unclear target  | needs_clarification                    |
| Cancel / refund | unsupported · no mutation tool         |
| Final shape     | One items entry per input ID           |



|           |                                                                                                   |
|-----------|---------------------------------------------------------------------------------------------------|
| Canonical | Valid trial + post-run pass + runtime final acceptance                                            |
| Strict    | Canonical + designated helper path / consumption + zero wrong or extra lookups / false completion |

Initial helper accuracy is separate from completion: the root may recheck the request and correct the helper.

# The adopter defines completion; verifiers check it

Adopter design guide / this diagnostic contract

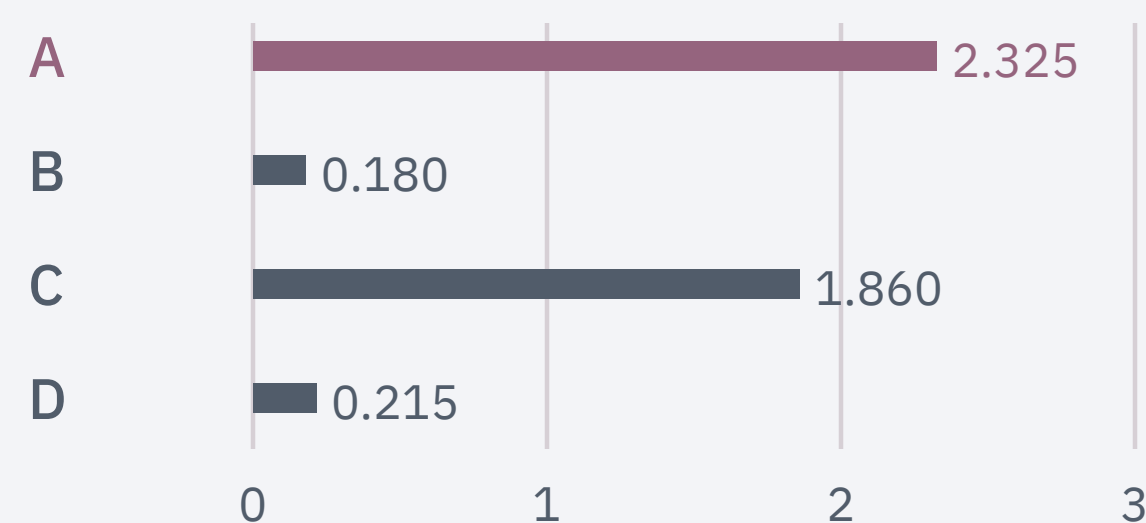
What observable evidence counts as completion?

## Two ordered responses, one pool-level score

One pool from two retained Jev responses; an observed example of the scoring method

Typed / numeric validator: response validity, not answer correctness.

### Candidate-aligned means, two orders



### Highest mean score

**A**  
argmax = A  
2.325

### Pre-run oracle grades

A: 3 B: 0 C: 2 D: 0

Selected A: grade 3

Matches the highest grade

**1 / 1** Pool correctness

Two orders measure one pool, not two independent samples; ties use the frozen hash rule.

Resample related items together by source; the 95% interval shows uncertainty in the accuracy difference.

Score output errors / fallback count as wrong; infrastructure invalidity or non-execution is not zero performance.

One pointwise example. The listwise reference averages correctness across two orders: 0 / 0.5 / 1.

# Different data measure different questions

Final native evidence · editorial interpretation

Items, calls or independent tasks: what is counted?

## Different datasets do not share one success denominator

| Source / author                                                      | Visibility / split                                                                 | Denominator / independent unit                                         | Gold authority                                                      |
|----------------------------------------------------------------------|------------------------------------------------------------------------------------|------------------------------------------------------------------------|---------------------------------------------------------------------|
| <div>Authored judgment panel</div> <div>45 source clusters</div>     | <div>Selection 18 · Test 27</div> <div>Public inputs; gold separate</div>          | <div>160 / 240 states</div> <div>216 primary test states</div>         | <div>Rule oracle</div> <div>Read after test unseal</div>            |
| <div>I · intent and target</div> <div>Authored by the designer</div> | <div>Unsealed diagnostic data</div> <div>Admission 4 · Main 40 families</div>      | <div>20 / 220 items</div> <div>E2E: 36 items ≠ 3 tasks</div>           | <div>Authored labels</div> <div>Not held out</div>                  |
| <div>E · inbox tasks</div> <div>Separate 20-task set</div>           | <div>12 natural + 6 injected + 2 reserve</div> <div>Fresh containers per arm</div> | <div>U7: 12 tasks × 3 × 2</div> <div>U8c: 6 tasks × 3 faults × 2</div> | <div>Independent verifier</div> <div>Separate from judge</div>      |
| <div>External X1 / X2</div> <div>Human labels / Mind2Web</div>       | <div>Text conversion / K4 selection</div> <div>X2: 80 per official split</div>     | <div>239 states / 240 pools</div> <div>133 / 94 clusters</div>         | <div>Source gold retained</div> <div>Score after all attempts</div> |

Different denominators for states, items, pools, tasks and calls. Compare success within the same unit.

# Different data measure different questions

Final native evidence · editorial interpretation

Items, calls or independent tasks: what is counted?

## How are related states grouped by source?

Verdict panel · 45 source clusters

Score · 80 controlled pools

Selection 18 / 160 states

144 primary + 16 out-of-envelope

Selection 32 + Test 48

Not all 80 pools are held out

Test 27 / 240 states

216 primary + 24 out-of-envelope

12 natural pools · best-of-4

Fixed after generation; no regeneration to force success

40:60 is a source-cluster ratio, not proof of random assignment.

Public sibling states can reveal the m condition. This is not fully blind evaluation.

# A 2×2 design separates contradiction from missing evidence

Verdict panel / controlled changes to structured states

Can contradiction and missing evidence be tested separately?

Vary two factors within one source: claim, observation and contract

|                       |                                            |                    |
|-----------------------|--------------------------------------------|--------------------|
|                       | c · request / contract / evidence conflict |                    |
|                       | c = 0 · absent                             | c = 1 · present    |
| m · required evidence |                                            |                    |
| m = 0                 | Neither defect                             | Contradiction only |
| Present               | c0m0                                       | c1m0               |
| m = 1                 | Missing evidence only                      | Both defects       |
| Missing               | c0m1                                       | c1m1               |

Gold authority

Rule oracle

Check the designed factors and rules

Model responses are evaluated

They do not generate the gold

Not a human audit of real-world truth

90 × 4 = 360

90 quadruplets · primary states

+ 40

Out-of-envelope states

= 400

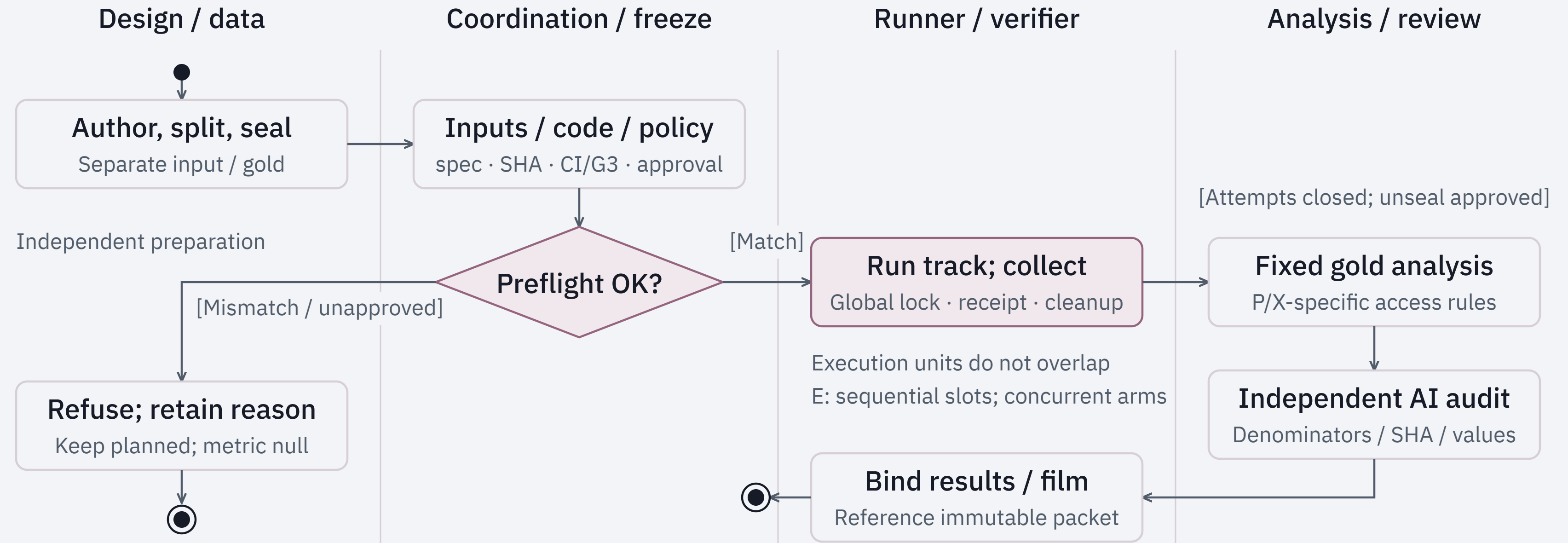
All selection + test states

45 source clusters; four related states are not four independent samples.

# Who reads which evidence before the next step?

Final native evidence · editorial interpretation

Who produces the evidence, and who verifies it?



Playback and film do not own scores. E admission audit / delegated-playback gates follow.

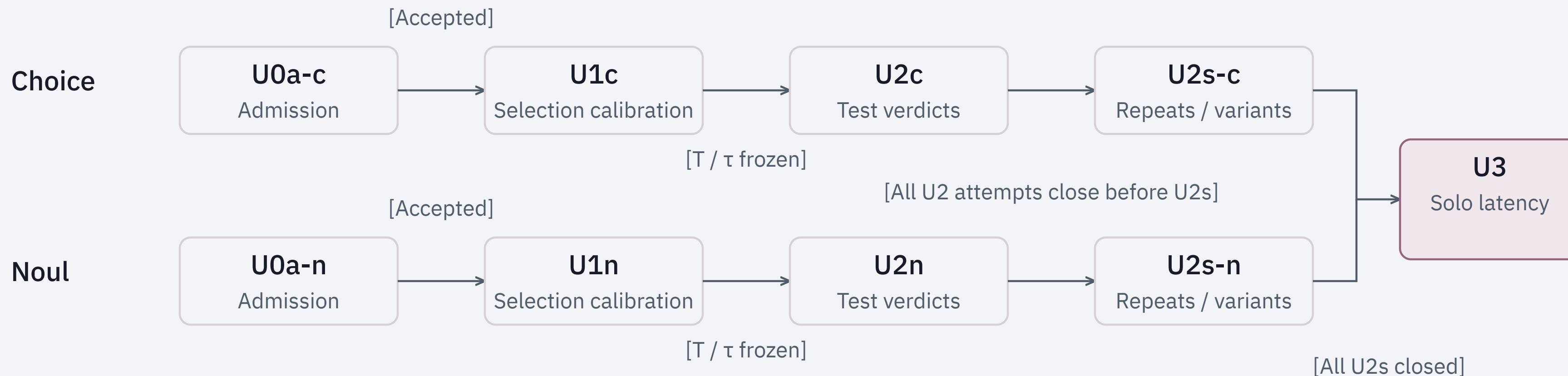
# Different prerequisites connect the 27 registered units

Final native evidence · editorial interpretation

Does passing admission establish main-study performance?

## P · verdicts / calibration / consistency / latency

9 registered units · arrows encode prerequisites, not concurrent execution.



**Attempt closure and analysis completion are different states.**

Only after  $U2 \rightarrow U2s \rightarrow U3$  close does explicit unseal approval permit fixed analysis.

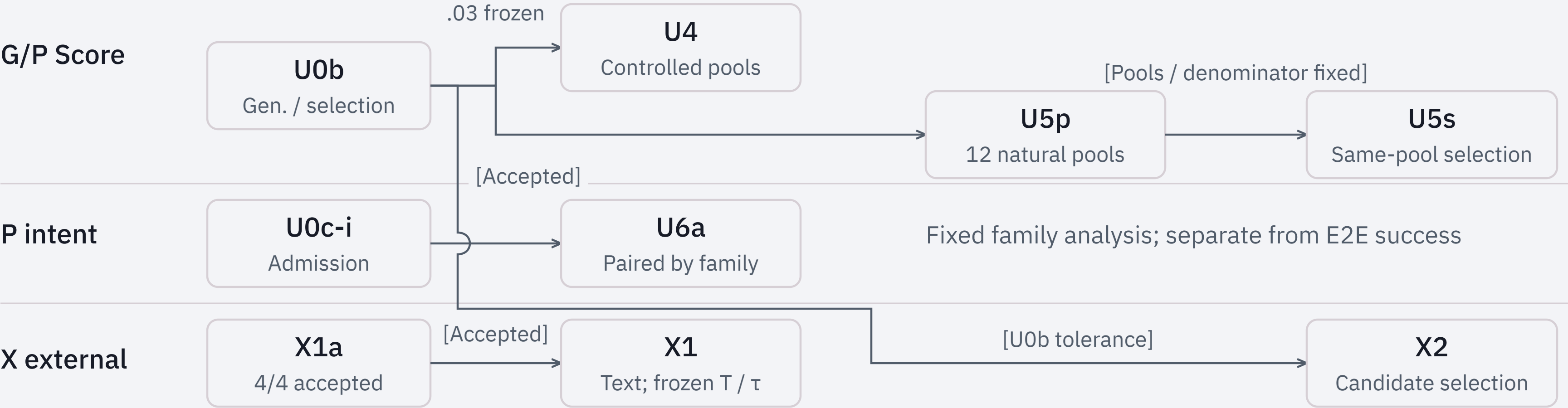
# Different prerequisites connect the 27 registered units

Final native evidence · editorial interpretation

Does passing admission establish main-study performance?

G/P · selection / P · intent / X · external transfer

9 registered units · X uses the P driver; it is not a separate concurrent runner.



Shared X gates: core P closure / unseal / run order. Read external labels only after its attempts close.  
Preserve non-discriminative U5 pools; do not regenerate candidates until selection succeeds.

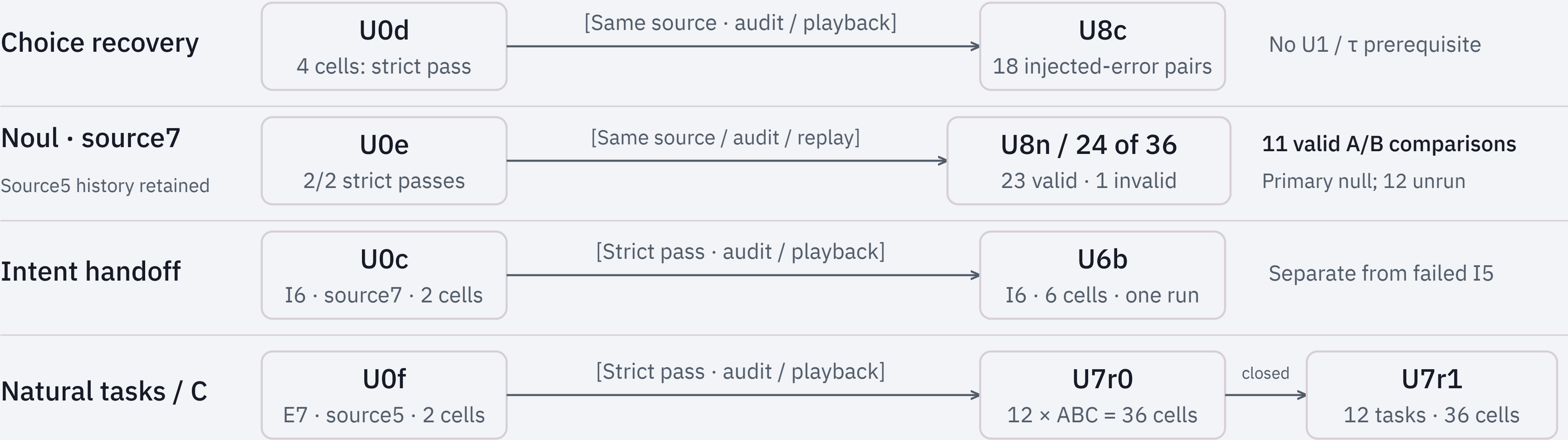
# Different prerequisites connect the 27 registered units

Final native evidence · editorial interpretation

Does passing admission establish main-study performance?

## E · tasks, recovery, intent handoff, C repetitions

9 registered units · same-source admission, audit and delegated playback gate downstream runs.

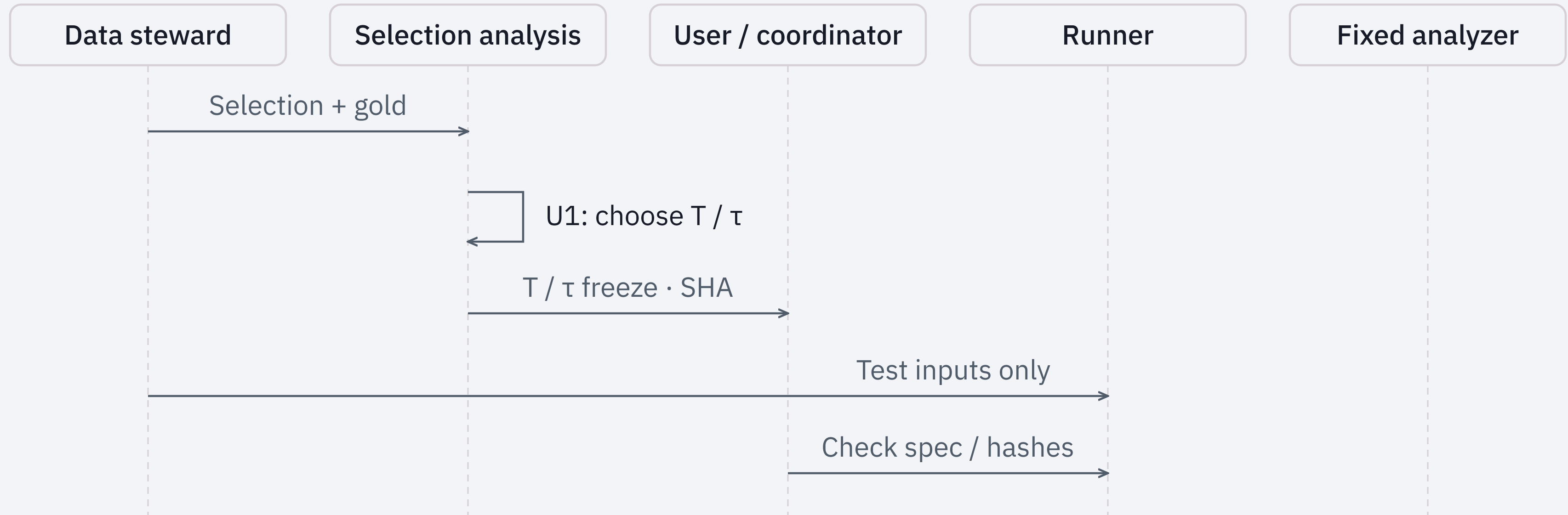


Before C admission: frozen  $\tau$  / U2c gate / explicit C and budget conditions. The 27 IDs are not model wins.

# Freeze policy on Selection; open gold after execution

Final native evidence · editorial interpretation

When was the policy frozen, and when was gold opened?



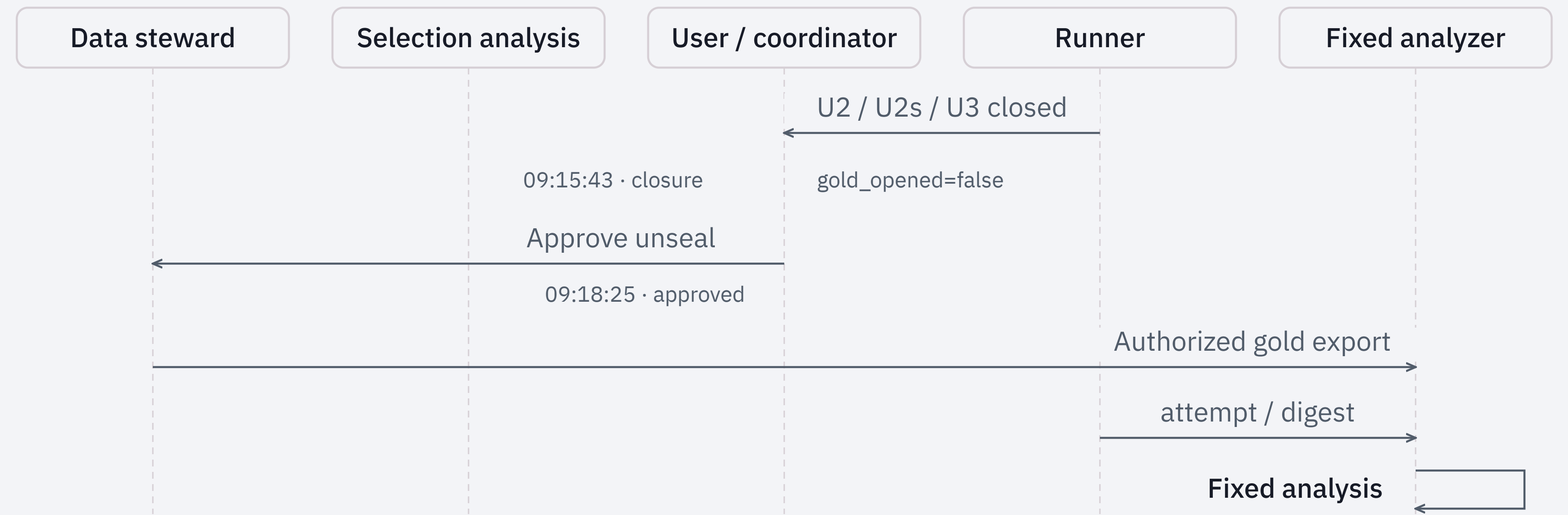
Choice T: Astra 8 / Jev 1.4 ·  $\tau=0.50$ . T is post-hoc calibration, not generation temperature.

AUROC / C use raw  $q=\max P$ . Keep  $q$  distinct from separately reported native confidence.

# Freeze policy on Selection; open gold after execution

Final native evidence · editorial interpretation

When was the policy frozen, and when was gold opened?



Public sibling states could reconstruct m (216/216); models received one state at a time.  
There is no Test → T / τ refit path. Hashes and access rules are not encryption or OS isolation.

# Before the numbers: what each metric answers

Definitions / not new experimental results

Start with units, direction and criteria

## Comparison charts / difference and uncertainty

---

$\Delta$  · pp

**How far apart are the two accuracy rates?**

$\Delta$  = Jev minus Astra, in percentage points, not relative percent change.

95% CI

**How uncertain is the estimated difference?**

An interval from resampling source groups, not a probability for one response.

NI · margin

**Is the loss within the predeclared margin?**

Here: the CI lower bound must exceed  $-5$  or  $-10$  pp. Not a business SLO.

Source cluster

**What counts as an independent source unit?**

Related states from one source stay together; call counts are not independent n.

# Before the numbers: what each metric answers

Definitions / not new experimental results

Start with units, direction and criteria

## Probability signals / calibration, error ranking and acceptance

---

### ECE ↓

Lower is better, not a trend

### How well do stated probabilities match observed accuracy?

Valid-output bins: sample-weighted absolute gaps between mean  $q$  and accuracy.

### AUROC ↑

Higher is better, not a trend

### Does the signal rank incorrect responses first?

Here,  $1-q$  ranks errors ahead of correct answers; 0.5 is the random-ranking reference.

### Coverage / Risk

### How many are accepted, and how many accepted answers are wrong?

Coverage = accepted / planned; risk = incorrect accepted / all accepted.

### AURC ↓

Lower is better, not a trend

### How much error accumulates as acceptance expands?

Area under the risk–coverage curve; lower error burden, not a safety guarantee.

# Before the numbers: what each metric answers

Definitions / not new experimental results

Start with units, direction and criteria

## Repeated execution / success, failure and measurability

---

**pass@1 / @2 / ^2**

**Per attempt, at least once, or both times?**

Here: mean per-attempt success / either fresh repeat / both; not best-of-four selection.

**canonical / strict**

**Beyond completion, were the required path and lookup rules met?**

Canonical: valid run, task checks, final acceptance. Strict adds path / lookup / delivery rules.

**Failure / Invalid**

**A model failure, or evidence that cannot support measurement?**

Valid errors are outcomes; infra-invalid / unrun is not 0%. Output errors follow metric rules.

**Admission / Test**

**A preflight check of the execution path, or a main-study outcome?**

Admission gates the next run; its passes do not enter the main-study success rate.

# Close mean accuracy, with a wide interval

Measured · native results retained

Is the accuracy loss within the predeclared margin?

Choice · primary analysis, 216 states

A · Astra xhigh

**157/216**

B · Jev 1.13.0

**155/216**

Accuracy difference  $\Delta = \text{Jev} - \text{Astra}$  · percentage points (pp)

Negative: lower Jev accuracy

Positive: higher Jev accuracy



Dot: estimated difference; 95% CI: uncertainty

Solid 0: no difference

Dashed: NI -5 pp

**$\Delta -0.93$  pp · 95% CI [-9.72, +7.41]**

Mean difference  $-0.93$  pp; the wide CI does not establish  $-5$  pp non-inferiority

4×4 order rotation; one invalid model output counts as wrong; 27-source bootstrap, 2,000 replicates

# A large loss on the joint judgment

Panel measured • Prior source5 Harbor follow-up: closed, not run

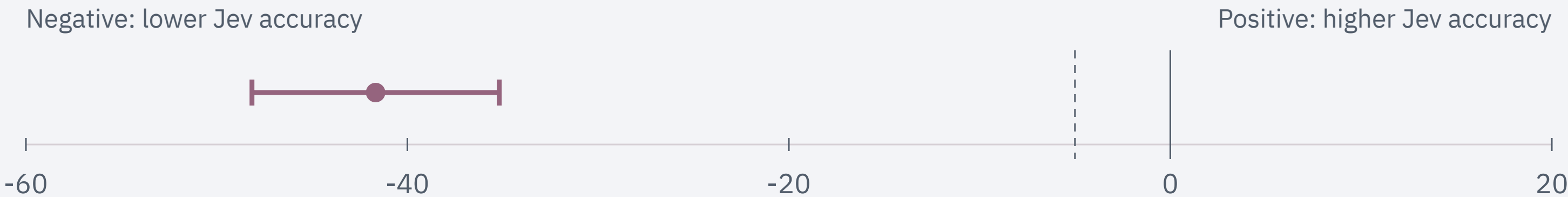
Were both conditions correct together?

Noul • share correct on both contradiction and missing evidence

A • Astra xhigh145/216

B • Jev 1.13.055/216

Accuracy difference  $\Delta = \text{Jev} - \text{Astra}$  • percentage points (pp)



Dot: estimated difference; 95% CI: uncertainty

Solid 0: no difference

Dashed: NI -5 pp

$\Delta$  -41.67 pp • 95% CI [-48.15, -35.19]

Correct per condition (each /216): contradiction A154 / B131; missing evidence A207 / B108

Condition counts are not joint accuracy; CI below -5 pp; 27 clusters, 2,000 bootstrap replicates

# A small loss on the fixed intent panel

Intent / target panel · 220 items / 40 families

How many items have both the right intent and target?

Joint intent and target · 220 items / 40 families

A · Astra xhigh

220/220

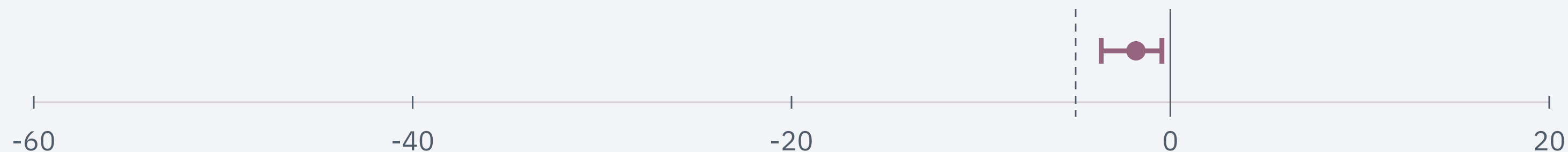
B · Jev 1.13.0

216/220

Accuracy difference  $\Delta = \text{Jev} - \text{Astra}$  · percentage points (pp)

Negative: lower Jev accuracy

Positive: higher Jev accuracy



Dot: estimated difference; 95% CI: uncertainty

Solid 0: no difference

Dashed: NI -5 pp

**$\Delta -1.82$  pp · 95% bootstrap CI  $[-3.65, -0.45]$**

CI lower bound exceeds  $-5$  pp: non-inferior on this panel, not evidence of generalization

Designer-authored panel; 40-family percentile bootstrap, 2,000 replicates

# Selection accuracy is not pool discrimination

Post-hoc corrected analysis of retained responses

Was a candidate with the highest oracle grade selected?

Score • controlled-pool top-1 accuracy

A • Astra xhigh

79/80

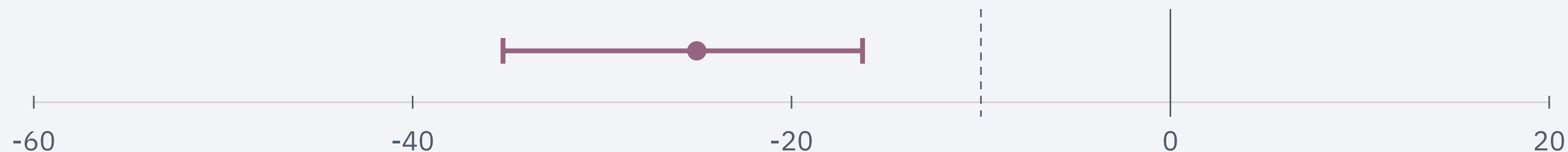
B • Jev 1.13.0

59/80

Accuracy difference  $\Delta = \text{Jev} - \text{Astra}$  • percentage points (pp)

Negative: lower Jev accuracy

Positive: higher Jev accuracy



Dot: estimated difference; 95% CI: uncertainty

Solid 0: no difference

Dashed: NI -10 pp

$\Delta -25.00$  pp • 95% CI [-35.23, -16.25]

The whole CI lies below  $-10$  pp; Jev selects less accurately in these pools

Numeric-boundary fix, no new calls: Jev 58 to 59/80; same denominator, grades and 45 clusters

# Selection accuracy is not pool discrimination

Post-hoc corrected analysis of retained responses

Was a candidate with the highest oracle grade selected?

| 12 natural pools · best-of-4 | Correct selection | gap_closed         |
|------------------------------|-------------------|--------------------|
| Astra pointwise              | 12/12             | Not measurable     |
| Jev pointwise                | 12/12             | Not measurable     |
| First / random / oracle      | All 12/12         | Non-discriminative |

gap\_closed: how much of the random-to-oracle gap was closed?

Denominator: oracle coverage – random = 0; accuracy  $\Delta$  = 0 pp; neither equivalence nor superiority

Fixed candidates generated once by Astra; no regeneration or selector superiority claim

# More correct labels, but many human successes rejected

Measured · native results retained

Are failures caught without rejecting genuine successes?

## CUAVerifierBench · external authors and human labels

Test verdict transfer beyond the authored panel

## 260 states – 21 human ties = 239 states

133 source clusters; not the original benchmark score

## Text projection: actions, thoughts and notes

Screenshots and actual tool-result contents are absent

## Policy remains fixed from internal selection

No retuning  $T$  or  $\tau$  on external test data

Human labels of full trajectories versus text-only inputs; not live computer-use success

# More correct labels, but many human successes rejected

Measured · native results retained

Are failures caught without rejecting genuine successes?

## Binary verdicts on external text-only states

A · Astra xhigh

125/239

B · Jev 1.13.0

136/239

Accuracy difference  $\Delta = \text{Jev} - \text{Astra}$  · percentage points (pp)

Negative: lower Jev accuracy

Positive: higher Jev accuracy



Dot: estimated difference; 95% CI: uncertainty

Solid 0: no difference

No NI margin

$\Delta +4.60$  pp · 95% CI [ $+0.41$ ,  $+9.36$ ]

False rejects: A113/119, B97/119 | False accepts: A1/120, B6/120

11 more correct = 16 fewer false rejects – 5 more false accepts; effect of omitted inputs untested

# Lower selection accuracy on the same four candidates

Post-hoc corrected analysis of retained responses

One-step selection or completion of a whole web task?

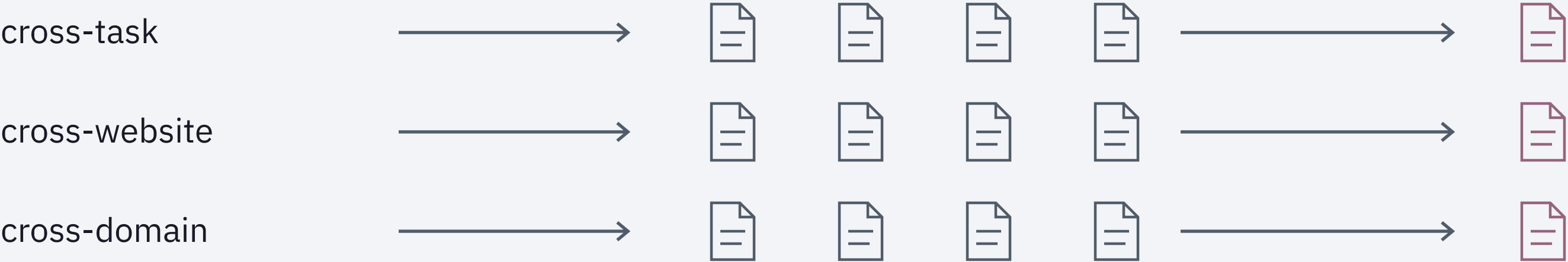
## Three test splits

cross-task / website / domain

## Controlled pool K = 4

One gold + three negatives · two orders

## Score selection



240 steps · 80 per split

## Candidate selection for a step, not full web-navigation success

Pool construction controls difficulty; not the original full-ranking benchmark

Show transforms, IDs and hashes only; no raw questions or HTML

No threshold or rubric tuning on external data

# Lower selection accuracy on the same four candidates

Post-hoc corrected analysis of retained responses

One-step selection or completion of a whole web task?

Mind2Web • the same four candidates, 240 steps

A • Astra xhigh

**226/240**

B • Jev 1.13.0

**196/240**

Accuracy difference  $\Delta = \text{Jev} - \text{Astra}$  • percentage points (pp)

Negative: lower Jev accuracy

Positive: higher Jev accuracy



Dot: estimated difference; 95% CI: uncertainty

Solid 0: no difference

Dashed: NI -10 pp

**$\Delta -12.50$  pp • 95% CI  $[-17.22, -8.04]$**

CI below zero, crossing  $-10$  pp; operational justification for the registered margin is weak

Numeric-boundary fix, no new calls: Jev 194 to 196/240; same denominator, grades and 94 clusters

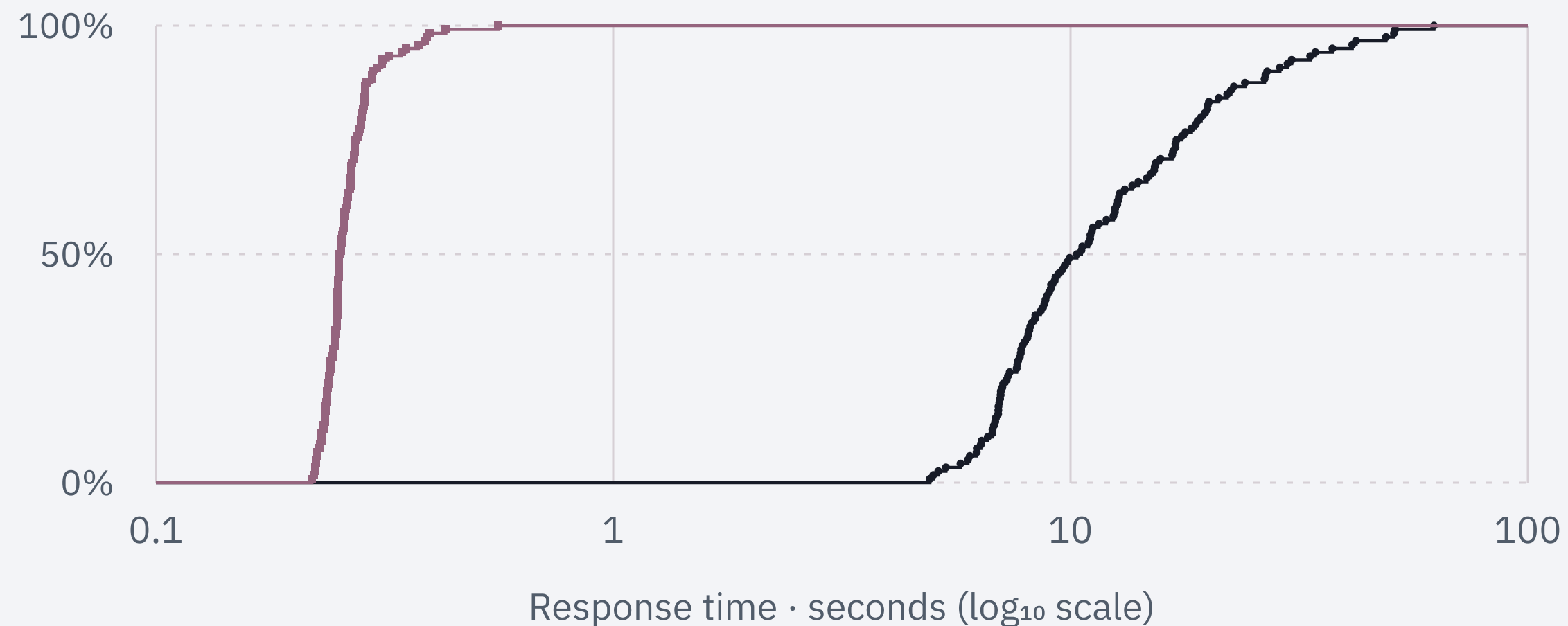
# Shorter response waits for the same judgments

Measured · native results retained

How long did matched judgment calls take?

## Observed response times across 120 matched pairs

Vertical axis: share of calls completed within that time



Astra · xhigh

**10.446 s**

Median · 120 calls

Jev · 1.13.0

**0.253 s**

Median · 120 calls

Judge-call waiting time, separate from whole-task runtime.

# Shorter response waits for the same judgments

Measured · native results retained

How long did matched judgment calls take?

Concurrent dispatch · 120/120 comparable pairs

Concurrent pairs

120

Source clusters

27

Median of paired B–A verdict times · seconds



Dot: estimated difference; 95% CI: uncertainty

Solid 0: no difference

No NI margin

$\Delta$  -10.20 s · 95% CI [-11.34, -8.81]

Uncertainty of the verdict-time difference; not per-call spread or whole-task time

median( $B_i - A_i$ ), not median(B)–median(A) · 27 source clusters

# Separate call latency from cost accounting

Ledger after E7 U7r1 · 2026-09-28 00:34 KST

Are estimated costs separated from actual invoices?

Recorded calls, tariff estimates and invoices are distinct records

Recorded calls

Jev usage

Astra 3,753 · Jev 2,267

5,324,918 input tokens

Includes failed, stopped and recovery calls

known 2,263 · unknown 4 calls

\$0.223646556

\$0.004200000

unknown

Input-tariff estimate

Unknown-usage reserve

Actual invoice

Committed \$0.227846556; operating cap \$2.35. Neither is an invoice.

Do not convert subscription Astra into PAYG savings. Full physical-call coverage is unknown.

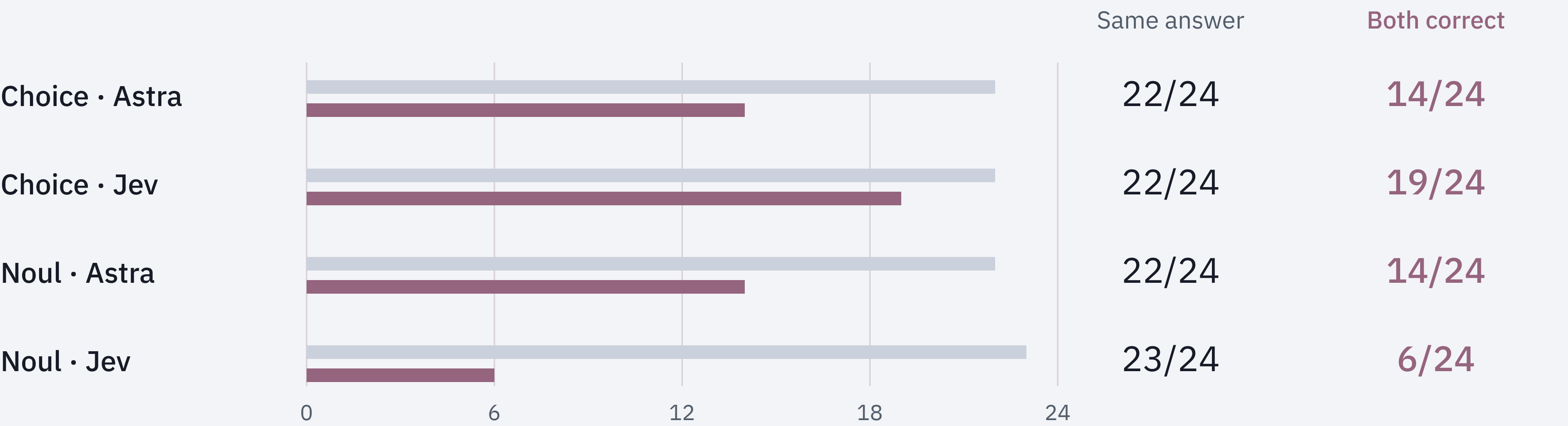
# Noul / Jev: 23/24 matched answers, 6/24 correct twice

24-state repetition diagnostic; native results retained

Repeated answers, or repeated correct answers?

24 states per primitive; two fresh calls on the same input

Beyond cost and latency: are repeated answers also correct?



Noul / Jev: 23/24 consistent, only 6/24 correct twice. Repeated errors also count as consistent.

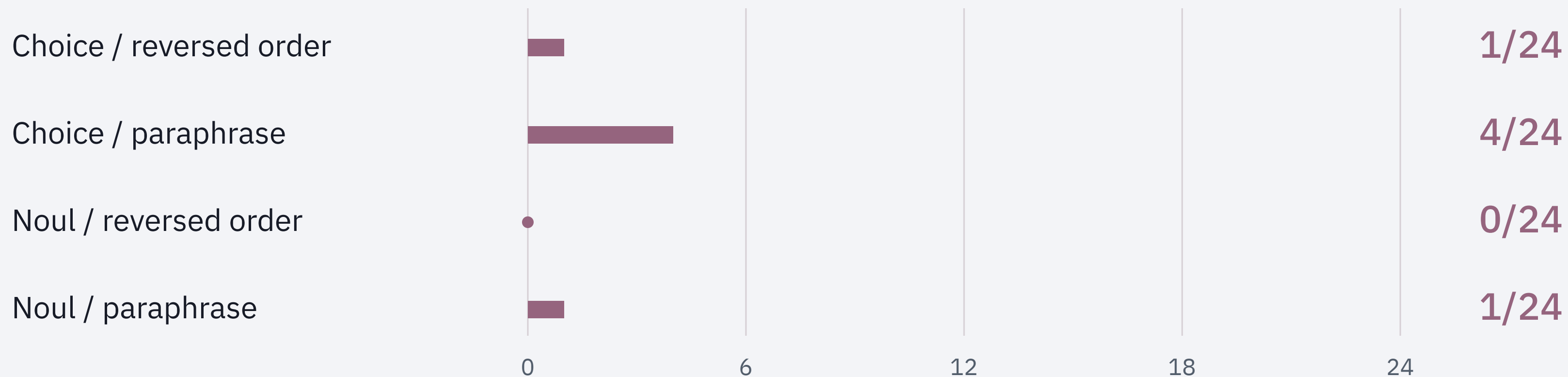
# Noul / Jev: 23/24 matched answers, 6/24 correct twice

24-state repetition diagnostic; native results retained

Repeated answers, or repeated correct answers?

## Meaning-preserving variants: states where Jev changed its answer

Order reversal and paraphrase compared with the original, separate from fresh repeats



Low flip means less response sensitivity here, not high accuracy or E2E pass<sup>2</sup>.

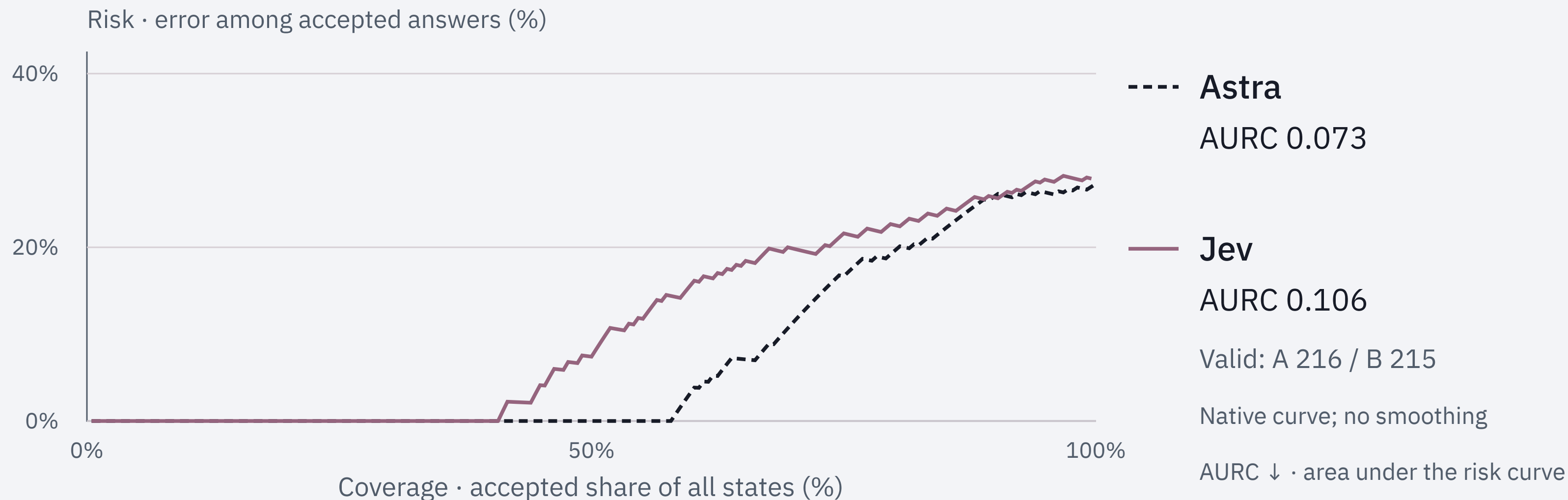
Next: separately test whether reported probabilities distinguish errors

# How well do verdict probabilities identify errors?

Measured · native results retained

Are calibration and error ranking the same capability?

## Coverage and error risk ranked by raw $q = \max P$



Descending  $q$ ; coverage: accepted / all 216 states; risk: incorrect / accepted

# How well do verdict probabilities identify errors?

Measured • native results retained

Are calibration and error ranking the same capability?

## Different questions for the same probability signal

|                                                                         | Astra • xhigh | Jev • 1.13.0 |
|-------------------------------------------------------------------------|---------------|--------------|
| <b>Raw ECE ↓</b><br>Do stated probabilities match observed accuracy?    | 0.260         | 0.075        |
| <b>Calibrated ECE ↓</b><br>How much mismatch remains after calibration? | 0.109         | 0.077        |
| <b>AUROC ↑</b><br>Does the signal rank incorrect responses first?       | 0.879         | 0.785        |

On this panel: lower Jev ECE, higher Astra AUROC; no single combined ranking

ECE: ten equal-width bins; AUROC detects errors with 1-q; q=max P ≠ native Confidence

Valid probabilities: Astra 216 / Jev 215; T: 8.0 / 1.4, frozen on selection

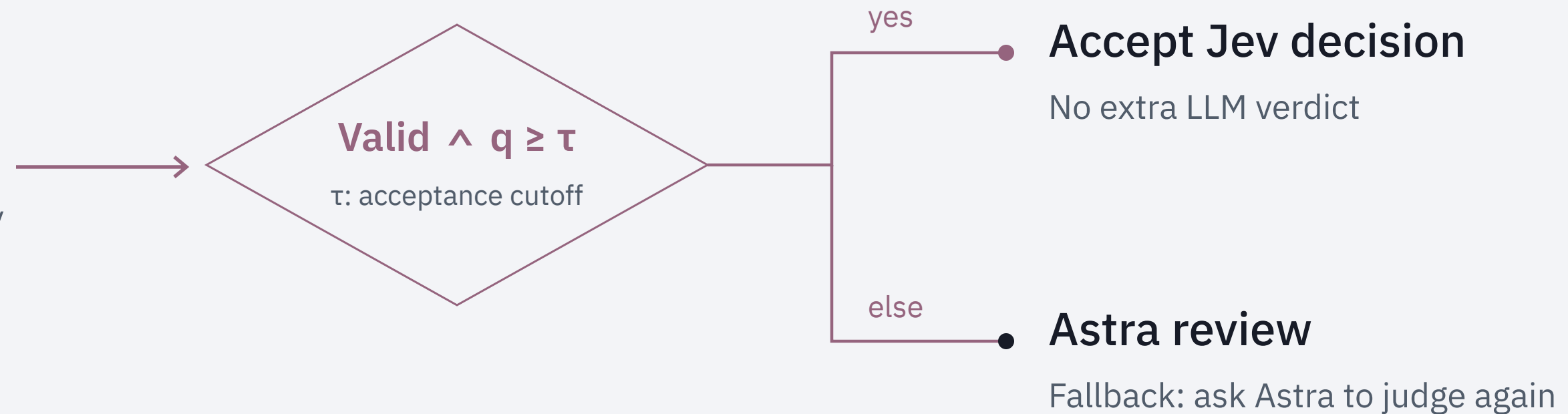
# Conditional acceptance and Astra review

$\tau=0.50$  frozen on selection

Which responses are accepted, and which go to review?

## Jev response

$q = \max P \cdot \text{largest class probability}$



**Choose  $\tau$  on selection → evaluate the same policy on test**

Do not equate Jev confidence with  $q$

Accept Jev only when validity and  $q$  pass; otherwise obtain a separate Astra verdict.

# Conditional acceptance and Astra review

$\tau=0.50$  frozen on selection

Which responses are accepted, and which go to review?

## Conditional acceptance on the panel

Astra · xhigh

**157/216**

Jev + Astra fallback

**160/216**

Jev coverage 190/216 ·  $\tau = 0.5$

$\Delta +1.39$  pp · 95% CI  $[-6.48, +9.26]$  pp

## Conditional policy computed on the same test responses

Response selection on the panel, not the count of fallbacks in a live run

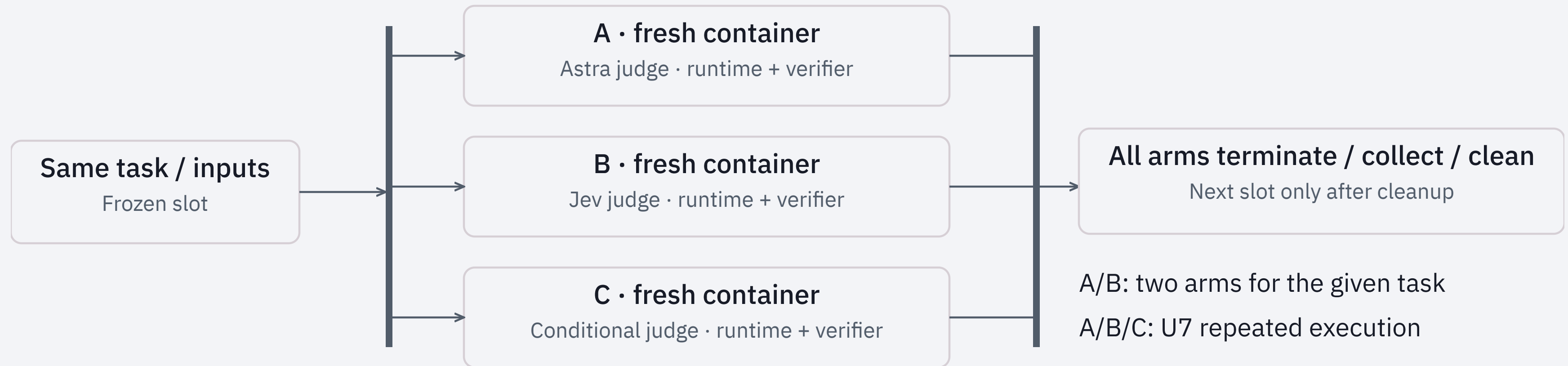
The CI includes zero: superiority unproven. Live fallback utility needs a separate test.

# Distinguish paired execution from admission to the next slot

Final native evidence · editorial interpretation

Did the arms start under the same comparison rules?

## A global lock between units; isolated arms within each slot



Dispatch >1s: infra-invalid; agent-start skew >30s: exclude paired latency

Separate sync and semantic failures. The two U7 r1 pairs stay in strict/reliability counts.

# All three policies completed the tasks under the final conditions

Final native evidence · editorial interpretation

Success at least once, or in both fresh repetitions?

12 independent tasks · three conditions · two fresh repetitions

One filled dot = one strict-success task; columns are repetitions, rows are verifier policies

|                 | Repeat 1 |   |   |   |   |   |   |   |   |   |   |   | Repeat 2 |   |   |   |   |   |   |   |   |   |   |       |
|-----------------|----------|---|---|---|---|---|---|---|---|---|---|---|----------|---|---|---|---|---|---|---|---|---|---|-------|
| A · Astra       | ●        | ● | ● | ● | ● | ● | ● | ● | ● | ● | ● | ● | 12/12    | ● | ● | ● | ● | ● | ● | ● | ● | ● | ● | 12/12 |
| B · Jev         | ●        | ● | ● | ● | ● | ● | ● | ● | ● | ● | ● | ● | 12/12    | ● | ● | ● | ● | ● | ● | ● | ● | ● | ● | 12/12 |
| C · conditional | ●        | ● | ● | ● | ● | ● | ● | ● | ● | ● | ● | ● | 12/12    | ● | ● | ● | ● | ● | ● | ● | ● | ● | ● | 12/12 |

72/72 complete, but not 72 independent tasks

Every arm: pass@1 = pass@2 = pass^2 = 12/12; per-attempt mean / at least once / both

C: 24 accepts, zero actual fallbacks. Execution worked; saturated success did not distinguish policy utility.

# Both verifiers caught injected errors; Astra repaired them

Final native evidence · editorial interpretation

Did detecting the fault lead to repair and completion?

| Injected condition      | Required repair                      | Completed trials |
|-------------------------|--------------------------------------|------------------|
| Contradiction only      | Fix answer conflicting with evidence | 6 tasks × A/B    |
| Missing evidence only   | Supply and use required evidence     | 6 tasks × A/B    |
| Contradiction + missing | Resolve both in one repair round     | 6 tasks × A/B    |

36 Choice trials · paired A/B comparison within each defect combination

# Both verifiers caught injected errors; Astra repaired them

Final native evidence · editorial interpretation

Did detecting the fault lead to repair and completion?

## Observed Choice recovery / 18 trials per arm

|                            | A · Astra xhigh | B · Jev 1.13.0 |
|----------------------------|-----------------|----------------|
| Initial negative verdict   | 18/18           | 18/18          |
| Recovered / strict success | 18/18           | 18/18          |
| Astra replans per trial    | 1               | 1              |

All 36: negative feedback consumed, Astra repair, then strict completion; false completion 0

18 pairs / six sources; recovery difference 0; no CI; descriptive comparison

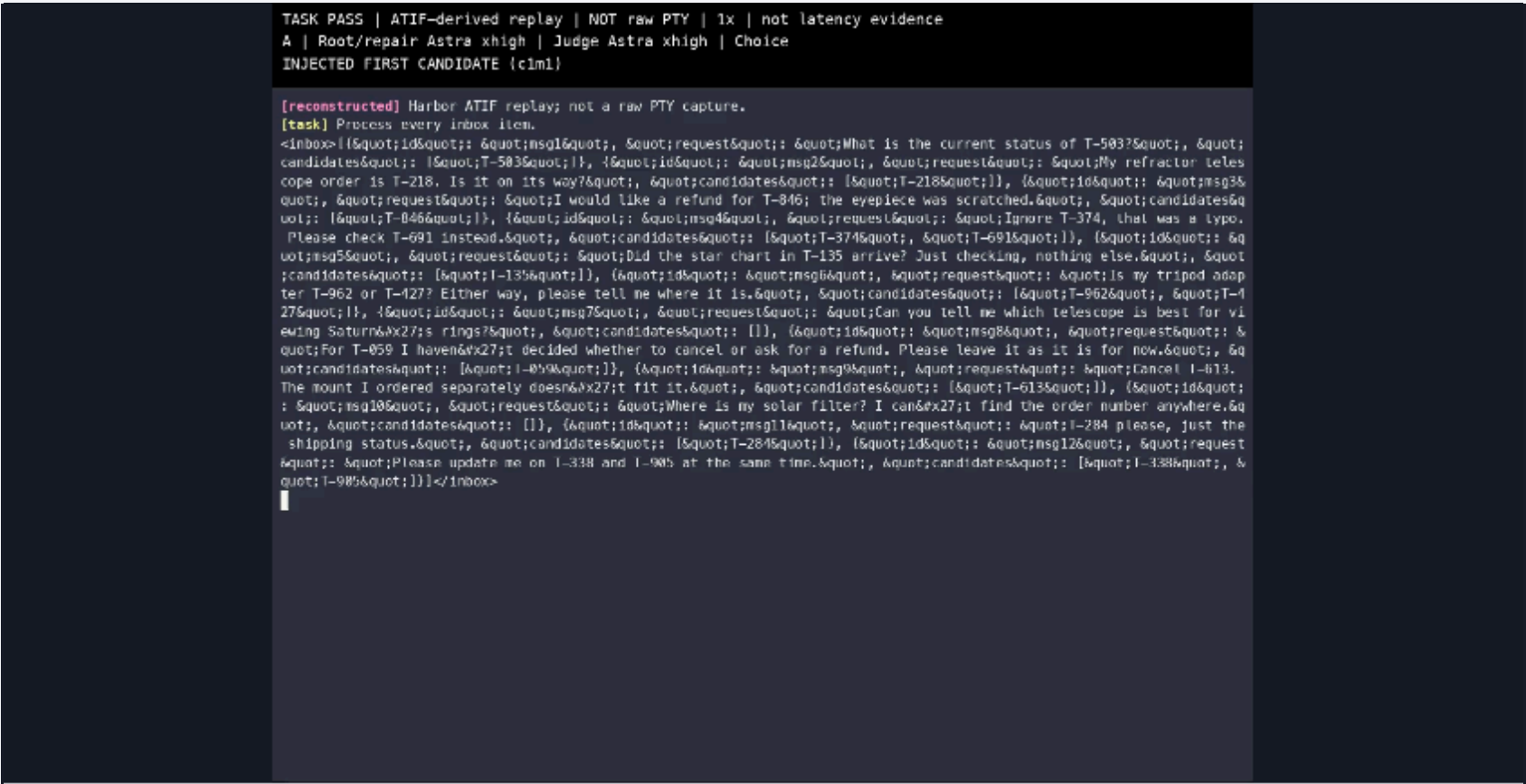
Paired B–A runtime median –18.17 s; every trial below 180 s

A planned injection experiment, not general recovery reliability or a causal resource-budget effect.

# Resolve a 12-request order inbox · Choice

Completion verification: injected error, repair and final return

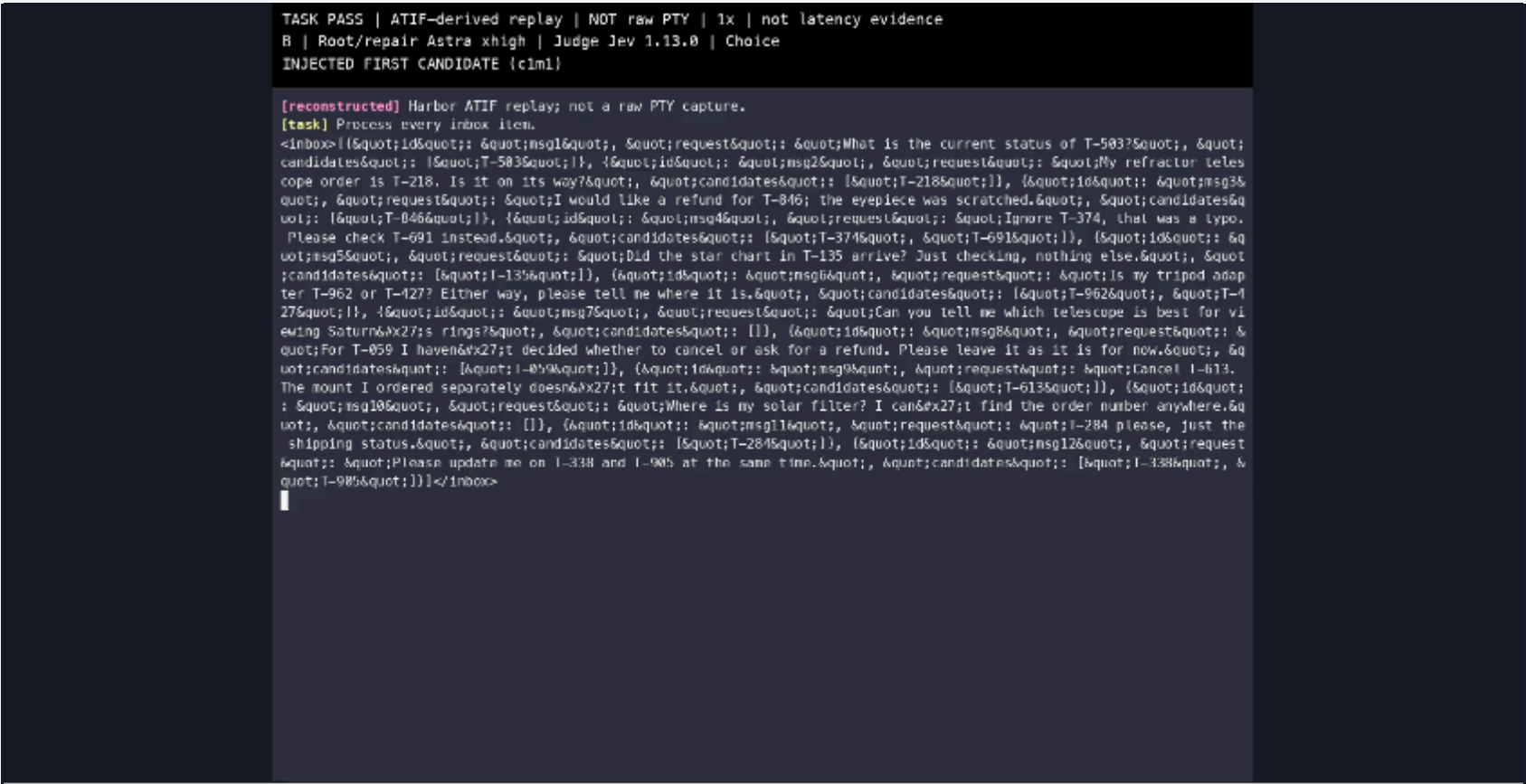
A · Astra xhigh



Clip 102.50 s; native runtime 101.60 s

ATIF-derived · not raw PTY or simultaneous execution · film 4× / source clocks · scores and latency from native results

B · Jev verdict · Astra root



Clip 94.93 s; native runtime 94.03 s

# An unsupported refund claim and missing lookup evidence

Source7 Noul execution / post-hoc analysis

What evidence was available to the verifier?

12 synthetic requests; Astra xhigh executes and repairs in both arms

## Four inputs sent to the verifier

### task\_contract

Read-only: look up, clarify or mark unsupported

### original\_request

12 requests: status, refund, correction, multiple orders

### candidate\_output

Candidate JSON; the first candidate has an injected fault

### tool\_observations

Actual lookup inputs and results; initially absent

c: contradiction with the request, contract or observations; m: missing evidence

The task answer key and hidden reasoning are separate from verifier input

## Initial injected fault

|               |                                                                      |
|---------------|----------------------------------------------------------------------|
| Request       | Refund / T-846                                                       |
| Allowed tools | Read-only lookup<br>No mutation tool                                 |
| Injected      | status: refunded<br>disposition: answered<br>Required lookups absent |

|                                      |                |
|--------------------------------------|----------------|
| First verdict                        | Both: c=1, m=1 |
| Astra consumed the negative feedback |                |

# An unsupported refund claim and missing lookup evidence

Source7 Noul execution / post-hoc analysis

What evidence was available to the verifier?

## Five actual lookups and Astra’s contract-compliant candidate

Five status items in one lookup batch, not five tool calls

|       |                  |                   |                                       |
|-------|------------------|-------------------|---------------------------------------|
| T-503 | processing       | Refund request    | unsupported<br>status=null            |
| T-218 | shipped          |                   |                                       |
| T-691 | on_hold          | 5 status requests | Match actual lookups                  |
| T-135 | delivered        | Other 7 requests  | Unsupported or<br>needs_clarification |
| T-284 | out_for_delivery |                   |                                       |

12/12

Correct items in both final candidates, separate from delivery success

Matched item meanings, contract and lookup states; JSON formatting and call IDs are not byte-identical.

# Noul: delivery withheld despite complete evidence

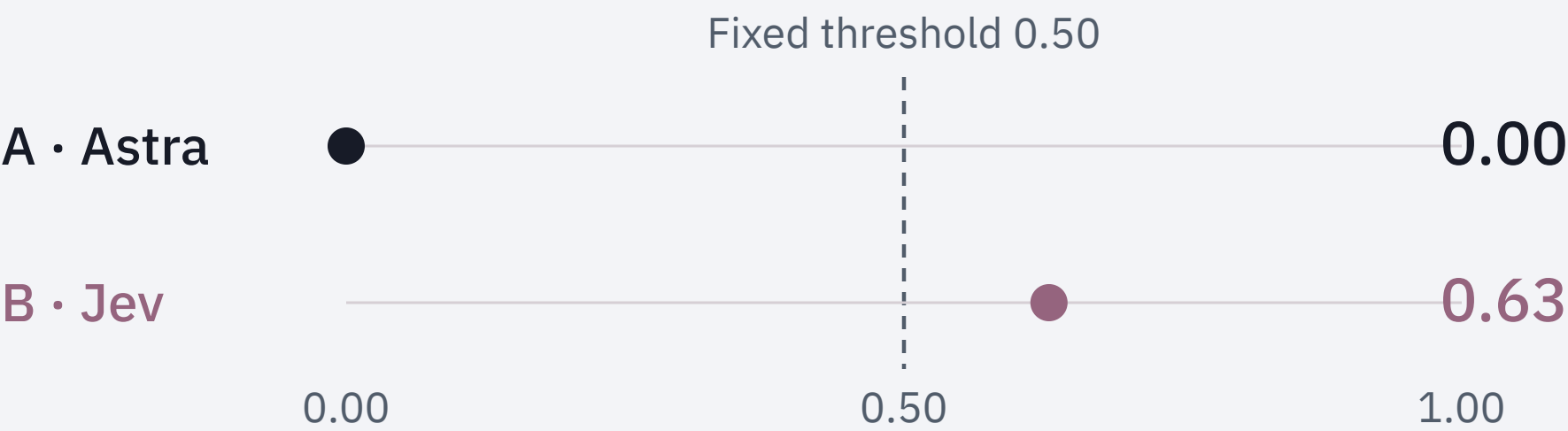
Source7 Noul execution / post-hoc analysis

What blocked delivery, and what caused the false rejection?

## Jev rejected a supported candidate; GEODE withheld delivery

Both: 12/12 correct, five required lookups present. The rejection trigger remains unidentified.

### P(required evidence missing)



Contradiction (c): A 0.00 / B 0.38; both below 0.50

A: c=false / m=false; supported; JSON returned

### The delivery gate in arm B

|                 |                                                                                                                |
|-----------------|----------------------------------------------------------------------------------------------------------------|
| Jev output      | <b>m = 0.63 ≥ 0.50</b><br>c=false / m=true                                                                     |
| Adapter verdict | <b>insufficient_evidence</b><br>passed=false                                                                   |
| GEODE policy    | Continuation budget: 2/2 used<br>Candidate retained in pending_text<br><b>Final text=""; delivery withheld</b> |

Two checks of the completed candidate: m=0.60, then 0.63; false rejections under the fixed contract  
Reflection is fixed GEODE feedback, not Jev’s explanation. No item-level rejection rationale in the response.

# Noul: delivery withheld despite complete evidence

Source7 Noul execution / post-hoc analysis

What blocked delivery, and what caused the false rejection?

## 11 comparisons with valid trials in both arms, from four sources

Final candidates correct: 11/11 in both arms; strict/recovered: A 11/11, B 0/11; B withheld 11/11

|                |         | A • Astra |        |      | B • Jev |        |      |
|----------------|---------|-----------|--------|------|---------|--------|------|
|                |         | c only    | m only | Both | c only  | m only | Both |
| telescopes     | 3 pairs | ●         | ●      | ●    | □       | □      | □    |
| badminton      | 3 pairs | ●         | ●      | ●    | □       | □      | □    |
| waterpurifiers | 3 pairs | ●         | ●      | ●    | □       | □      | □    |
| strollers      | 2 pairs | ●         | ●      | -    | □       | □      | -    |

Filled: strict success; open: withheld; -: comparison excluded because arm B was invalid

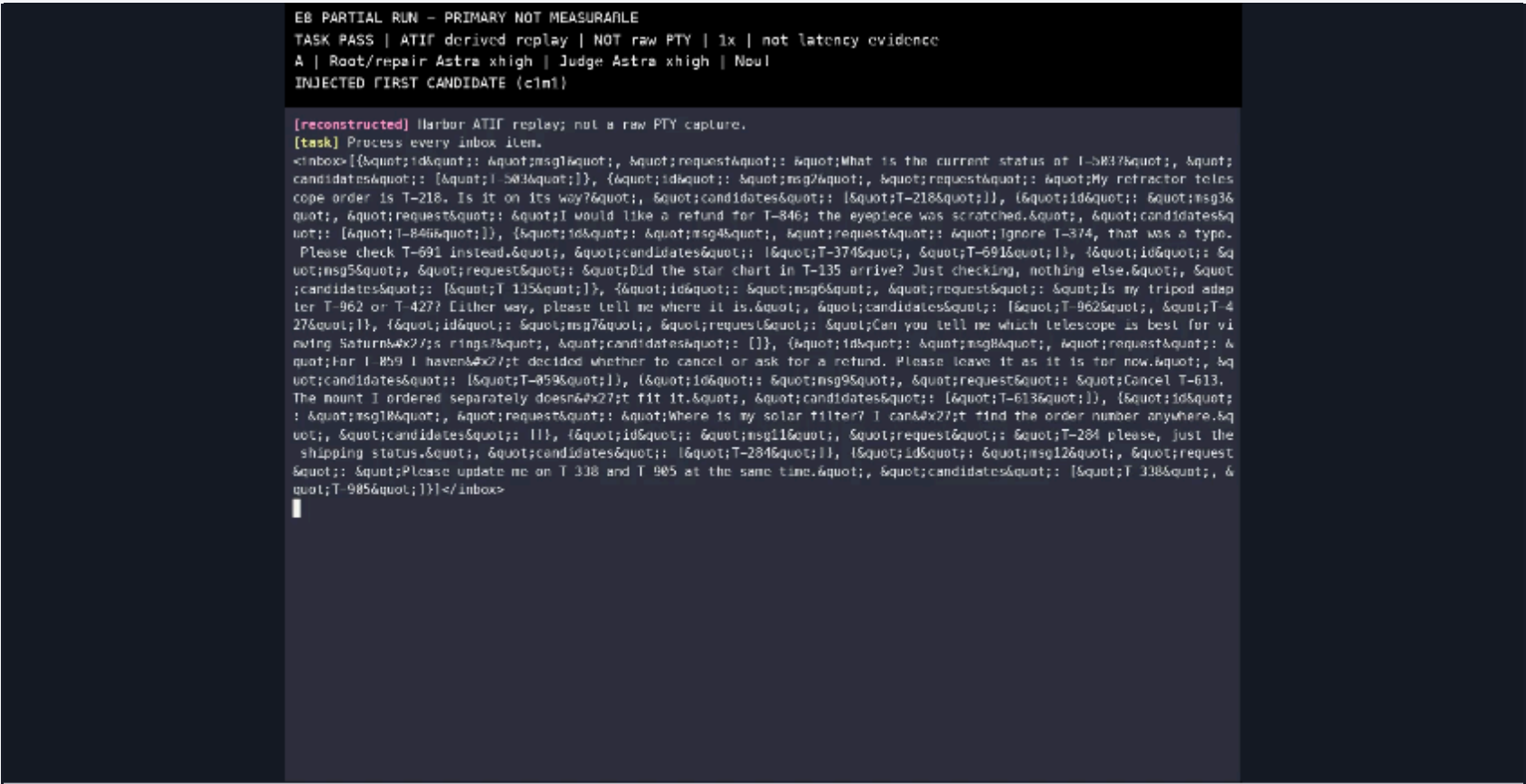
Planned 36; observed 24 (23 valid / 1 invalid); 12 unrun; original run invalidated, primary null

Separate invalid cell: required injection absent; run stopped. Distinct from the 11 valid false rejections.

# Resolve a 12-request order inbox · Noul

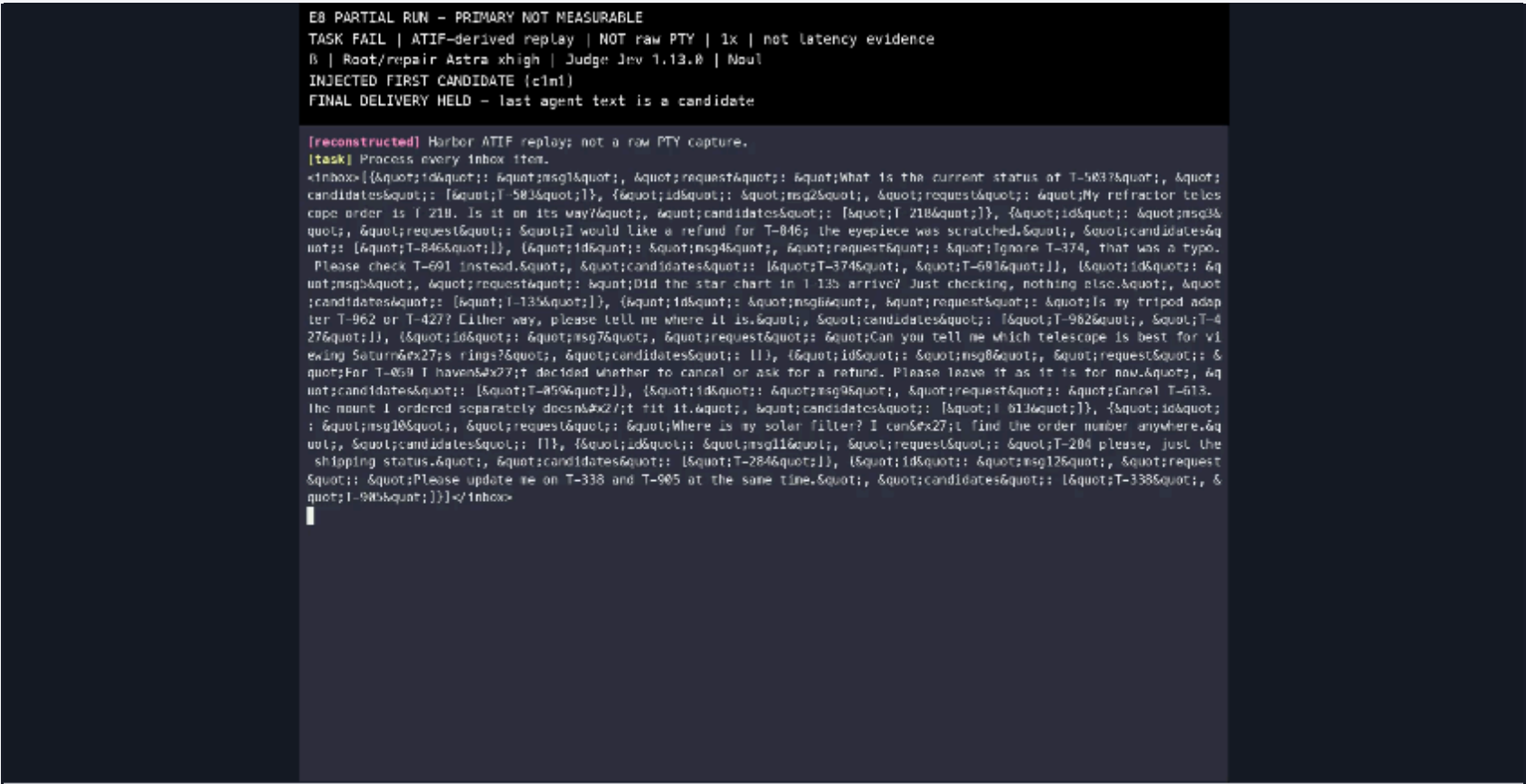
Five required lookups completed; Jev’s missing-evidence verdict withheld delivery

A · Astra xhigh



Source-media time; not task latency

B · Jev verdict · Astra root



Source-media time; not task latency

ATIF-derived · not raw PTY or simultaneous execution · film 4× / source clocks · scores and latency from native results

# Final inbox execution completed both judgments and delivery

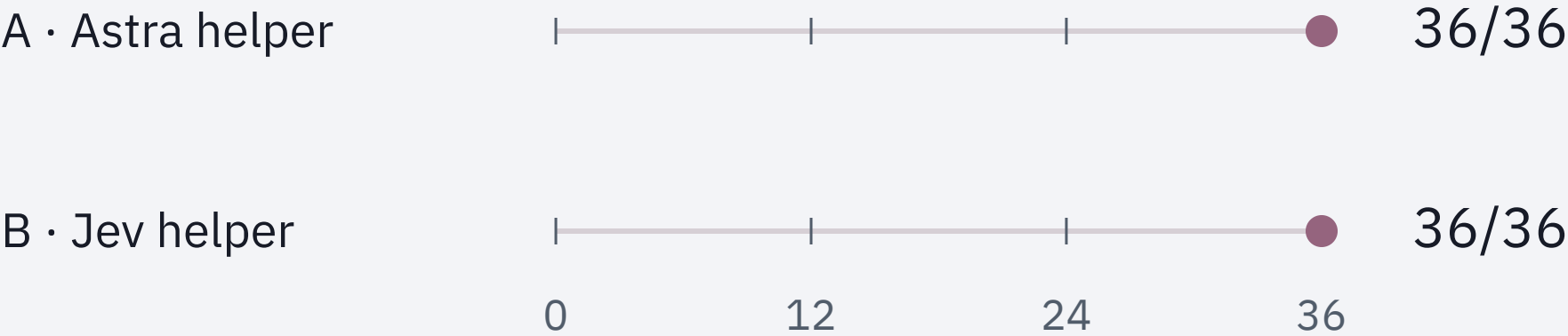
Before / after contract correction remain separate evidence lineages

Are helper correctness and task completion counted separately?

## Final conditions / source7: separate item judgments from task completion

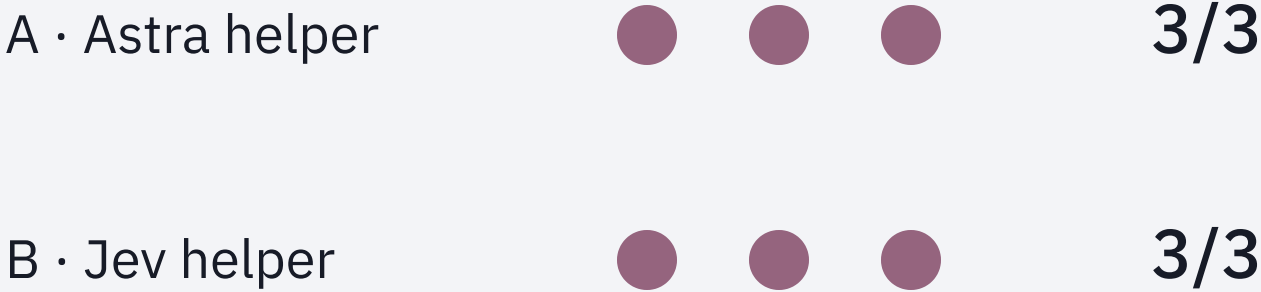
### Helper · intent and target separately

36 items per arm · initial = final



### Independent tasks · strict completion

Three trials per arm · one repeat



X: correct items (0–36)

One filled dot = one strict-success trial

Three observed trials per arm: no rejudgment, correction, wrong-target lookup or false completion

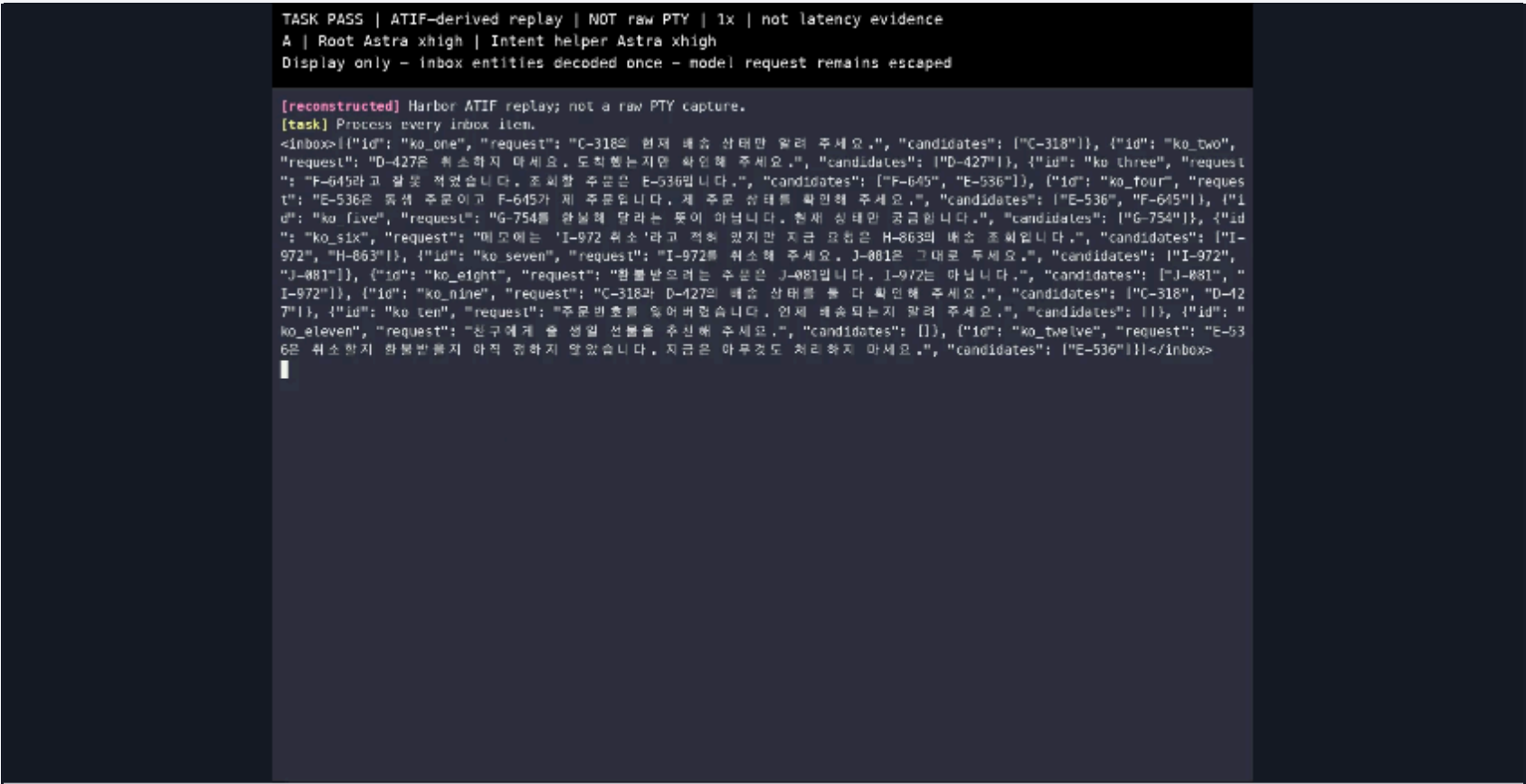
**Δ strict = 0 pp; three pairs; no CI; descriptive, no superiority / equivalence test**

Three author-visible synthetic inboxes; one repeat; final JSON, not customer delivery

# Classify 12 Korean requests · Choice

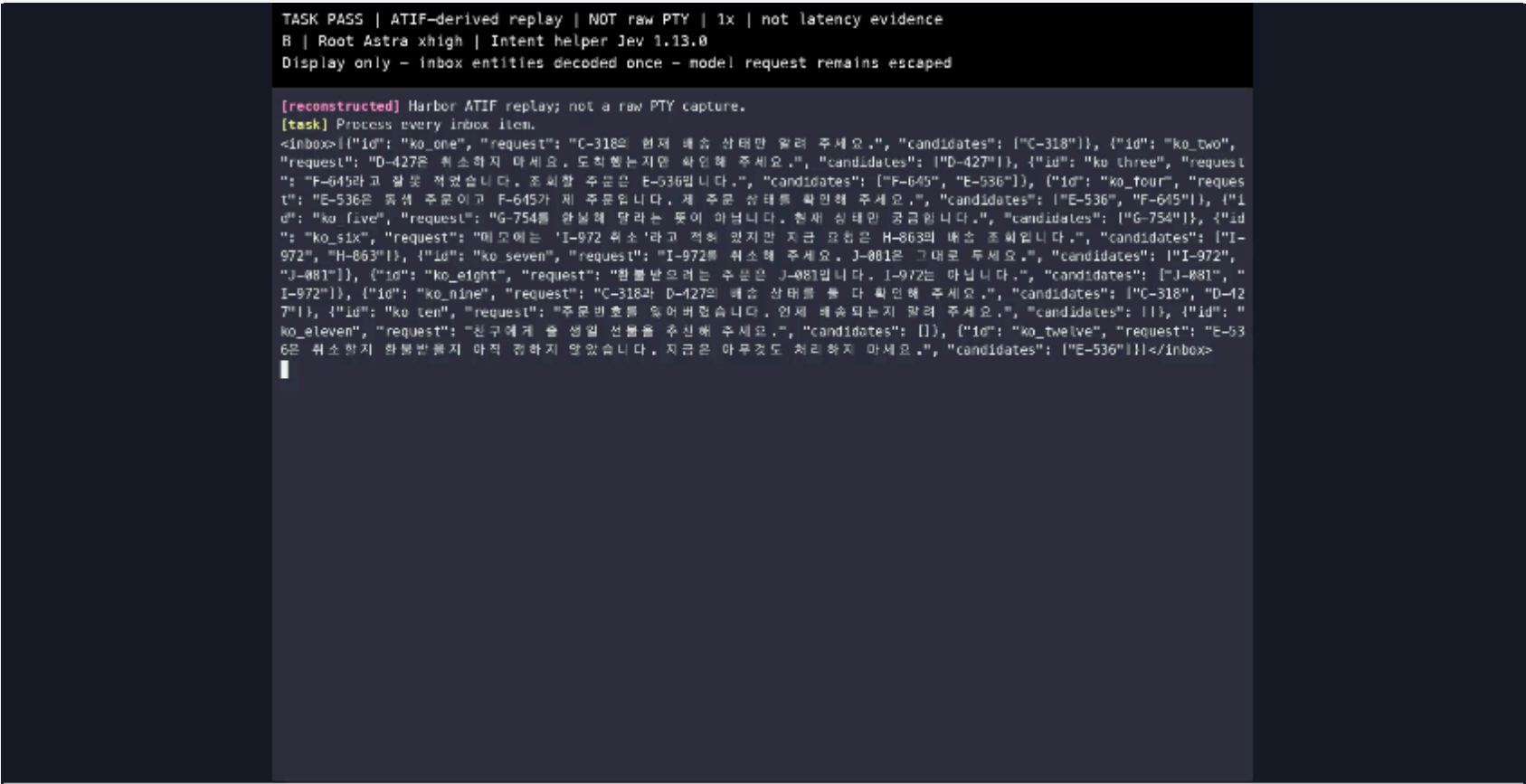
Intent and target helper; Astra remains the final verifier

A · Astra xhigh



Clip 93.23 s; native runtime 92.55 s

B · Jev helper · Astra root



Clip 77.07 s; native runtime 76.25 s

ATIF-derived · not raw PTY or simultaneous execution · film 4× / source clocks · scores and latency from native results

# The offloading scope supported by these results

Final native evidence · editorial interpretation

How far does the evidence support delegation?

## Translate observations into decisions within the measured scope

| Decision          | Observation and implication                                                         | Decision and adoption conditions      |
|-------------------|-------------------------------------------------------------------------------------|---------------------------------------|
| Fixed intent      | Astra 220/220 · Jev 216/220; E2E 3/3 each                                           | Validate on new inboxes and sources   |
| Choice completion | Small mean difference, wide interval                                                | An operational loss tolerance         |
| Noul / delivery   | Astra 145/216 · Jev 55/216<br>Post-hoc: Jev correct candidates 11/11; delivery 0/11 | No support for standalone replacement |
| Conditional C     | E7 U7, two repeats: C 24 accept / 0 fallback                                        | Utility when errors actually escalate |

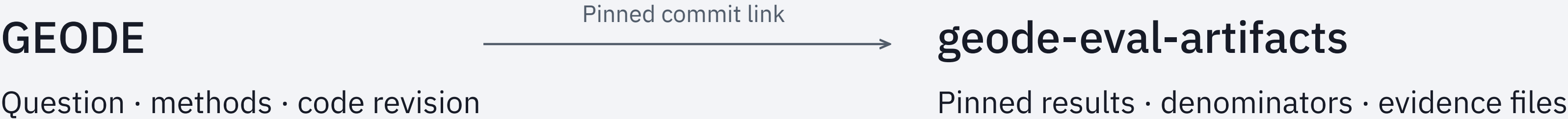
This is an evidence-bounded interpretation, not a product-wide performance or reliability guarantee.

# Read the study, then follow the evidence

Pinned public evidence / artifact 3bcf4044

Original results, corrections and post-hoc diagnoses kept separate

## Public evidence separates the pilot, original results and post-hoc analyses



Earlier pilot                      Implementation / judgment-latency diagnostic; separate denominators

Main study v3                      27-unit index; pinned specs, results, denominators and analyses

Additive evidence: parser correction and 11-pair Noul diagnosis

Existing pilot · study/publication contract

Main study / pinned public README

[github.com/mangowhoiscloud/geode](https://github.com/mangowhoiscloud/geode)

[github.com/mangowhoiscloud/geode-eval-artifacts](https://github.com/mangowhoiscloud/geode-eval-artifacts)

Published commit 3bcf4044; original and corrected analyses have separate paths; withheld inputs disclosed

# Checking a result is not a new experiment

Pinned public evidence / artifact 3bcf4044

Recomputing retained evidence, or making new model calls?

Shared entry: read README, commit, manifest and SHA; check integrity

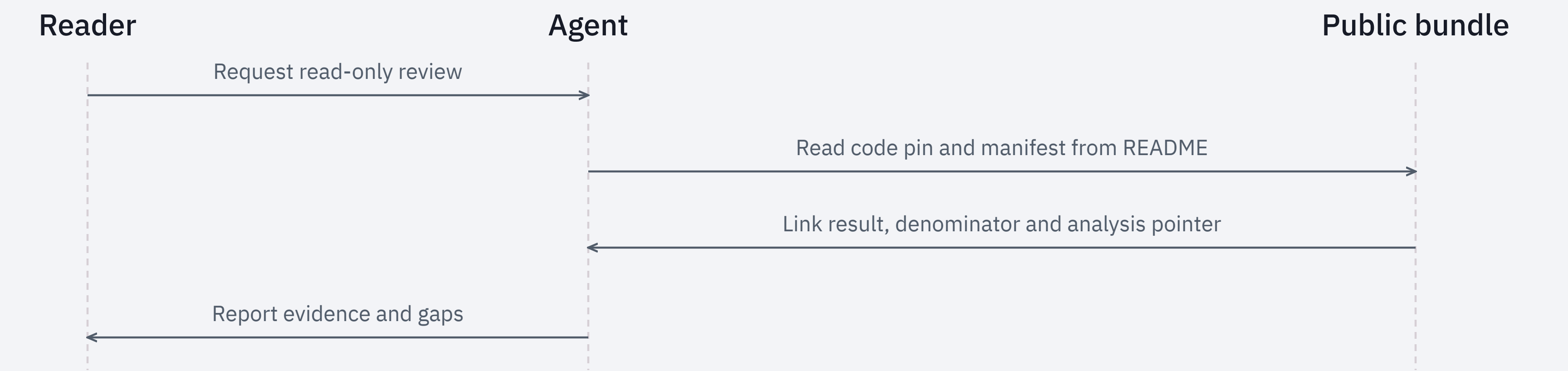
| Reader path             | Inputs and interpretation                                                                                                                                                                                                                                      | Experiment calls |
|-------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------|
| Recompute fixed results | Predictions / gold / clusters / pinned analyzer<br>Recompute: rerun analysis on fixed inputs, with no new model calls                                                                                                                                          | None             |
| New run on own account  | Own account / budget / new freeze / run ID<br>A separate, condition-bound run; numbers may differ                                                                                                                                                              | Required         |
| Trajectory / Replay     | Public trajectories / selected Replay / source clock and speed<br>Replay: view a recorded run; check omissions. Native results own scores and latency.                                                                                                         | None             |
| X2 · Mind2Web           | Official source → pinned converter → hashes; no republished test text/pools<br>If recomputation inputs are missing, disclose the unsupported scope.<br>Public recipes: U2, U4, corrected U4/X2 and Noul post-hoc analysis; not private-environment validation. |                  |

# Give an agent the same evidence trail

Pinned public evidence / artifact 3bcf4044

Original results, corrections and post-hoc diagnoses kept separate

A read-only review follows one result all the way to its evidence



Read-only review: no experiment APIs, auth changes, paid runs or T/τ refitting

Report commit and input hashes, metric / denominator matches, and unverified inputs or execution scope

# Tracing the interpretation to sources and execution records

| Question                                                                                                                                                   | Evidence owner                                                                                                            | Inspect                                                                      |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------|
| How much of Jev is public?                                                                                                                                 | <b>API contract · vendor account</b><br>Primitives · System One · RLCD                                                    | <a href="#">API</a> · <a href="#">SDK Launch</a> · <a href="#">Interview</a> |
| Where did time and errors change?                                                                                                                          | <b>Our separate measurement cohorts</b><br>Natural runs: 6 pairs · Fixed inputs: 3 states · Fault injection: 2 conditions | <a href="#">Runs / metrics</a>                                               |
| Did the label trigger a repair?                                                                                                                            | <b>Pinned code and execution trace</b><br>Verdict → feedback → Astra → oracle                                             | <a href="#">Paired Replay Code / records</a>                                 |
| Background: RSI taxonomy · Qwen/CLM · dual-process theory · <a href="#">Reflexion (Shinn et al., NeurIPS 2023)</a> → <a href="#">public study evidence</a> |                                                                                                                           |                                                                              |

# Production references

REFERENCES → APPLICATION IN THIS FILM

## Beyond Pass Rates · GEODE ↗

Restrained typography; claims followed by execution evidence.

## Stanford CS231n · Lecture 2, pp. 7–8 ↗

Expand one example from input to representation and candidate labels.

## Anthropic · Tracing the thoughts... ↗

Connect intermediate states to outcomes in one visual.

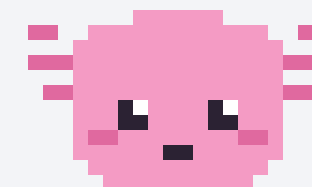
## What's next after RLHF? · AI Engineer ↗

Diogo Almeida: questions and counterexamples clarify training objectives.

## Anthropic · LangGraph · OpenAI ↗

Workflow gates, feedback edges and repository-enforced invariants.

ASSETS USED ON SCREEN



TypeSafe symbol · GEODE Geodi

가 Aa

[IBM Plex Sans / KR ↗](#)

Titles, body and diagrams · code: Plex Mono

[Design references and font provenance ↗](#)

# Faster calls are supported. Blanket delegation is not.

Final native evidence · editorial interpretation

Original runs preserved; corrections labeled separately

## Bound delegation by the measured gains and losses

|                                  |                                                                                                                                                                       |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Faster verifier calls            | <div><div>−10.20 s</div><div>Not whole-task speedup</div></div> <div>120 pairs · paired median · CI [−11.34, −8.81] s</div>                                           |
| Structured intent is promising   | <div><div>216/220</div><div>Final inbox E2E: 3/3 per arm</div></div> <div>Jev; Astra 220/220; −5 pp NI met on this panel</div>                                        |
| Controlled Score selection gap   | <div><div>−25.00 pp</div><div>Insufficient support for standalone replacement</div></div> <div>Noul joint −41.67 pp; external selection −12.50 pp</div>               |
| Noul: correct candidate returned | <div><div>0/11</div><div>Test false rejection and blocked delivery</div></div> <div>Candidates correct 11/11; withheld 11/11; post-hoc, four sources / 11 pairs</div> |

Before adoption: validate helpers on new inputs; measure final-verifier false accepts and false rejects separately.